The Rise of Alternatives*

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Abstract

Since the 2000s, U.S. public pension funds have actively shifted their risky investments away from public equities and toward alternative assets like private equity and hedge funds—some much more than others. We explore a range of possible explanations for these trends and argue that beliefs about the risk-adjusted return of alternatives have played a central role. Pension beliefs are shaped by investment consultants, peers, and experience in the 1990s. Other demand-side channels related to risk-seeking motives, spending needs, and fund governance have weaker empirical support. We also consider supply-side factors and agency interpretations of the data.

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Notes: Would be good to know what % of global assets were alternatives 2001 vs 2021

1 Introduction

Over the last few decades, public pensions in the United States have fundamentally changed how they take risk. At the turn of the century, risky investments—defined here as any holdings outside of fixed income and cash—were primarily in public equities. Alternative assets like private equity, real estate, and hedge funds accounted for just 14% of risky investments in 2001 but grew to 39% by 2021.[1] These national trends mask considerable heterogeneity across pensions. For example, the alternative-to-risky share for pensions in states like Maine, New Mexico, Indiana, Wyoming, and Texas has increased by an average of 58 percentage points since 2001. In contrast, the alternative-to-risky share for pensions in South Dakota, Nevada, Georgia, Iowa, and Colorado has hardly changed.

What has driven public pensions to actively reallocate their risky investments toward alternatives—and why have some done so much more than others? In this paper, we explore a range of possible explanations and conclude that beliefs about the risk-adjusted return of alternatives have played a key role. Other channels related to risk-seeking motives, spending needs, and governance appear weaker in the data. Some of the aggregate trend may also reflect the development of the private capital industry, though these supply-side forces are less likely to account for the wide variation in alternative use across pensions. Agency-based explanations that point to the illiquidity of alternatives and their insulation from mark-to-market accounting are also plausible, and we assess their ability to explain both the aggregate and cross-section in detail.

To structure our empirical analysis, we look to canonical mean-variance theory for guidance on the potential factors behind the rise of alternatives. We focus on this class of models because they remain widely used by pensions as an input to asset allocation decisions: 85 out of the top 100 largest U.S. pensions explicitly reference mean-variance optimization in their annual reports, asset-liability studies, or documents available on their websites. In textbook mean-variance models (Markowitz, 1952; Merton, 1969), the composition of the risky portfolio—summarized in our context by the alternative-to-risky share—is determined entirely by beliefs about expected returns and covariances. Thus, one natural explanation for the rise of alternatives is that pensions have become more optimistic about the so-called alpha of alternatives relative to public equities, some more than others. Formally, alpha is defined as: $\alpha = \mu_A - \beta\mu_E$, where μ_A is the expected excess return of alternatives over cash, μ_E is the expected excess return of public equities, and β is the expected covariance between the alternatives and public equities, scaled by the expected variance of

[1] Unless otherwise noted, we analyze target portfolio shares rather than actual portfolio shares to focus on active portfolio decisions. See Section 2.1 for additional background and discussion of target versus actual shares.

public equities.

A second explanation operates through increased risk tolerance. In the standard model, absent a shift in beliefs, any factor that raises investor risk tolerance (e.g., underfunding) will cause the overall risky share to increase but leave the alternative-to-risky share unchanged. This result underlies Tobin's celebrated two-fund separation theorem (Tobin, 1958). However, this logic breaks down if the investor faces a binding constraint on its total risky share, such as a leverage constraint. When risk tolerance increases but the constraint binds, the investor can no longer raise its total risky share and must instead reallocate within the risky portfolio. In this case, the alternative-to-risky share may rise, provided the investor is initially bullish enough on alternatives relative to public equities. We refer to this mechanism as the constrained risk-taking channel.

Our empirical analysis investigates both mechanisms described above, starting with the belief-based channel. A first piece of evidence comes from capital market assumptions (CMAs) published by the general consultants that advise nearly all public pensions on asset allocation (Andonov et al., 2023). These CMAs embed consultant beliefs about the expected return and covariance of different asset classes, including alternatives (Couts et al., 2023). On average, consultant-reported beliefs about the alpha of alternatives have increased by 58 basis points (bps) since the early 2000s. This rise can be almost entirely attributed to an increase in the expected return of alternatives, whereas perceived diversification benefits have slightly declined during this period (i.e., μ_A and β have both increased).

More directly, in the cross-section, pensions tend to have a higher alternative-to-risky share when their general consultant holds a more optimistic view about the alpha of alternatives. This positive correlation could arise if consultants have a causal impact on their clients' portfolios or if consultants are chosen based on their stated beliefs. To isolate the causal component, we leverage the fact that consultant selection is partly driven by geographical proximity (Andonov et al., 2023). An instrumental variables design based on this fact indicates that widening disagreement among consultants accounts for about one-third of the increase in cross-sectional variation in the alternative-to-risky share since the 2000s. Consistent with Bayesian learning by pensions from their consultants (e.g., Anagol et al., 2021), we also find that the causal effect of consultants declines with proxies for internal investment capacity, like plan size.

Consultants are not the only channel through which pensions form beliefs about alternatives. Drawing on past research linking belief formation to experience (Malmendier and Nagel, 2016; Andonov and Rauh, 2022), we hypothesize that investors who were late to invest in the stock market during the 1990s came to

view public equities less favorably than those who invested earlier, as their experience was more heavily influenced by the bursting of the dot-com bubble. This episode was especially salient for U.S. pension because it was the first time many had meaningfully invested in the stock market (see Section 2.2). Consistent with this idea, states whose pensions invested later in equities during the 1990s subsequently increased their alternative-to-risky shares more during the 2000s. Entry timing alone accounts for nearly one-fifth of the cross-pension variation in changes in the target alternative-to-risky share from 2002 to 2021, even after controlling for the total increase in public equity exposure during the 1990s, funding levels, local regulation, and several measures of financial sophistication.

Motivated by research on belief formation in social networks (Bailey et al., 2018, 2022), we also test whether pension behavior is shaped by peers. We find that pensions tend to invest more heavily in alternatives when their geographically close peers do as well—an effect that persists even after accounting for shared consultants and herding incentives (Scharfstein and Stein, 1990). The pattern remains when peers are defined as nearby private-sector institutions, such as endowments and corporate pensions, which differ in regulation and governance and are thus less likely to be affected by unobserved common shocks (Angrist, 2014). This latter result supports the idea that pension beliefs have been influenced by the behavior of high-performing university endowments (Lerner et al., 2007, 2008). These institutions began adopting alternatives in the 1990s under the so-called Yale Endowment model pioneered by David Swensen, who was famously bullish on alternatives (Goetzmann and Oster, 2012).

Next, we evaluate the constrained risk-taking channel. As a reminder, the core premise of this channel is that pensions want to take more risk but are constrained from increasing their target risky share, prompting them to instead increase their alternative-to-risky share. If this mechanism is operant, then pensions that shifted more heavily to alternatives should also show signs of being more constrained. We test this hypothesis using two proxies for constraint slack. The first is the gap between each pension's actual and target share. This measure is motivated by the fact that actual allocations are typically allowed to fluctuate within a preset range around target before rebalancing is triggered (Harvey et al., 2025). Thus, a pension that wants to take on more risk but is unable to formally increase its target will likely keep its actual risky share persistently above target. The second proxy is based simply on the cash share, which should be low for a risk-constrained pension. In both cases, the cross-sectional relationship with alternative adoption appears weak.

As a further test of the constrained risk-taking channel, we examine whether pensions with stronger

incentives to take risk also invest more heavily in alternatives relative to public equities. We consider a wide range of proxies for risk-seeking motives, including those related to underfunding (Pennacchi and Rastad, 2011; Mohan and Zhang, 2014; Lu et al., 2019), pension accounting and fund aging (Andonov, Bauer, and Cremers, 2017), and nominal return targeting. Across all of these measures, we consistently find a weak link with the alternative-to-risky share in the cross-section of pensions, both in long-run changes and levels. For example, a simple cross-sectional regression of changes in the alternative-to-risky share between 2002 and 2021 on changes in pension funding yields an R^2 of 1%. A research design that exploits state-level variation in exposure to the Covid-19 pandemic also provides little indication that exogenous shocks to life expectancy affect the alternative-to-risky share. Taken together, the evidence suggests that risk-taking and leverage constraints are unlikely to account for the stark differences in how aggressively pensions have actively reallocated their risky assets toward alternatives.

After analyzing the cross-section, we turn to the question of why pensions have increased their alternative-to-risky share in aggregate. On its face, the weak cross-sectional link between risk-seeking motives and the alternative-to-risky share casts doubt on their ability to fully explain the aggregate trend. Still, one might worry that this result reflects measurement error in our proxies for risk aversion or constraint slack. To address this concern, we simulate a variant of the mean-variance models used by most U.S. pensions for portfolio construction (Merton, 1969; Campbell and Viceira, 2002). Critically, the model allows us to quantify the strength of the constrained risk-taking channel in aggregate without requiring precise measures of pension risk aversion or constraint slack. In each simulation, we draw beliefs about expected returns and covariances from a broad distribution and infer the level of risk aversion needed to match the aggregate pension portfolio in 2001. Holding beliefs fixed, we then attempt to match the observed aggregate shift toward alternatives by increasing risk aversion, while assuming that pensions are constrained from increasing their risky share beyond the level observed in 2021.

Strikingly, in virtually all simulations (99.6%), it is not possible to lower risk aversion enough to match the observed aggregate change in the alternative-to-risky share. The intuition for this result is simple: by revealed preference, the low alternative-to-risky share in 2001 implies that pensions were bullish on public equities compared to alternatives. Consequently, absent a change in beliefs, they would shift their risky investments toward public equities and not alternatives when a constraint on their risky share binds. To be clear, this simulation exercise does not rule out the possibility that U.S. public pensions are reaching for yield or are leveraged-constrained. On the contrary, the fact that the *risky* share has increased from 69% to

76% over the last twenty years is a clear indication that effective risk aversion has indeed declined. However, the mean-variance model highlights that a shift in beliefs is still necessary to explain the concurrent change in the *composition* of the risky portfolio, namely the increase in the alternative-to-risky share.

The simulated model can also be used to gauge how much of the aggregate increase in the alternative-to-risky share can be explained by beliefs. Specifically, the model implies that pension beliefs about alpha would need to rise by about 70 bps to fully explain the observed aggregated trend. According to our estimates, about 12% of this increase can be causally attributed to the fact that the consultants have become more bullish about alternatives since the early 2000s. Additionally, using common techniques from network economics (Leontief, 1986; Acemoglu et al., 2016; Herskovic et al., 2020), we show that amplification by the peer network can account for another 20% of the belief-shift needed to match the aggregate trend. Consistent with a broad shift in beliefs about alpha, we further show that U.S. public pensions, U.S. endowments, U.S. corporate pensions, and U.K. corporate pensions have all increased their alternative-to-risky shares since the 2000s, even as their risky shares have diverged sharply.

In the final part of the paper, we consider two additional explanations for the rise of alternatives. The first is related to the development of the private capital industry, which has arguably made it easier for pensions to access parts of the economy in which it was previously difficult to invest. While this type of supply-side explanation may help explain the aggregate trend, it is less compelling as an explanation for the wide cross-sectional variation in the alternative-to-risky share that persists today. A related hypothesis is that, as private markets have grown and matured, public pensions have increased their allocations to alternatives in an effort to track the investable global market portfolio (GMP). Though the global share of alternatives has indeed increased since the early 2000s, U.S. public pensions are now heavily overweight alternatives relative to the GMP.

A second potential explanation is that stated beliefs about alpha reflect deeper agency frictions. For example, consultants may publish CMAs that are bullish on alternatives because they have career or financial incentives to steer pensions toward these asset classes. Similarly, both pensions and consultants may prefer alternative assets because they are infrequently marked-to-market, which complicates risk-adjustment and can give the appearance of outperformance by smoothing returns (Lo, 2001; Jurek and Stafford, 2015; Couts et al., 2020; Stafford, 2022; Korteweg et al., 2024). These agency-based explanations are plausible and difficult to test rigorously using observational data. Nonetheless, our findings help characterize the form such frictions must take to match the patterns observed in the data.

To explain global trends in the alternative-to-risky share, the capacity or desire to smooth returns via alternatives would need to have grown stronger over time across a broad range of public and private institutions. One possibility is that institutional investors broadly value return-smoothing, and the maturation of alternatives has made it easier to act on that preference. This view could also help explain why pensions are heavily overweight alternatives relative to the GMP.

On its own, a broad expansion in investment products that smooth returns cannot explain why pensions differ so markedly in their adoption and use of alternatives. To rationalize the cross-section, return-smoothing motives (or capacity) must vary across pensions in ways orthogonal to funding, size, governance quality (Hochberg and Rauh, 2012; Andonov et al., 2018), and the other controls used in our regression tests. While unobserved agency frictions may play a role in overall alternative use, several empirical patterns are more naturally accounted for by genuine differences in beliefs. Most notably, pensions vary considerably in how they allocate *within* subcategories of alternatives that are similarly illiquid and infrequently marked-to-market (e.g., real estate versus private equity). Moreover, these allocation choices also closely track consultant beliefs.

The analysis in this paper adds to a large literature on institutional investor behavior. A recurring question in recent work is why investors appear to take more risk as interest rates fall (Rajan, 2005; Borio and Zhu, 2012; Stein, 2013). Explanations for this "reach-for-yield" behavior commonly invoke agency frictions (Becker and Ivashina, 2015), institutional constraints (Campbell and Sigalov, 2022), and behavioral biases (Lian, Ma, and Wang, 2018). In the case of U.S. public pensions, underfunding and accounting distortions have been cited as key drivers of the rising risky share since the early 2000s (Eaton and Nofsinger, 2004; Pennacchi and Rastad, 2011; Mohan and Zhang, 2014; Andonov et al., 2017; Lu et al., 2019).

In contrast to prior work, we ask why pensions have shifted the *composition* of their risky assets away from public equities and toward alternative assets over this period. This reallocation is substantial: since 2001, the alternative-to-risky share has increased by 25 pp, compared to just 8 pp for the risky share, with considerable heterogeneity across pensions. Importantly, this compositional shift also fully accounts for the rise in alternatives as a share of *total* assets (Ivashina and Lerner, 2018), given both the modest increase in the risky share and the low initial alternative-to-risky share.[2] Our central contribution is to offer a simple explanation for these patterns: pensions—some more than others—increasingly believe that alternatives

[2]From 2001 to 2021, the alternative share of total assets increased from 9% to 30%. The low alternative-to-risky share in 2001 (14%) and modest increase in the risky share (8 pp) imply that this increase was driven by a reallocation within the risky portfolio.

generate alpha relative to public equities. This belief-driven account contrasts with popular reach-for-yield narratives (Gillers, 2021), which help rationalize changes in the risky share, but have limited power to explain its composition. The mean-variance models used by most pensions reinforce our interpretation of the data. Absent an increase in perceived alpha, those models cannot easily generate an increase in the alternative-to-risky share, especially given how low that share was in 2001.

Our findings also connect to research emphasizing the role of beliefs in shaping household investment decisions (Pástor, 2000; Leombroni et al., 2020; Beutel and Weber, 2022; Giglio et al., 2021). Much like households, pension beliefs appear influenced by consultants (Foerster et al., 2017), peers (Bailey et al., 2022), and past experience (Malmendier and Nagel, 2016; Bordalo et al., 2022). In institutional settings, prior work shows that clients who share a specialized consultant tend to select similar active managers and adopt comparable benchmarking practices (Jenkinson et al., 2016; Andonov et al., 2023; Augustin et al., 2024; Martinez and Qian, 2024). We build on this work by showing that consultant beliefs—particularly their growing optimism about the expected return of alternatives—have had a causal impact on client portfolios.[3] We also extend the evidence of investor extrapolation from Andonov and Rauh (2022) by linking the rise of alternatives to pensions' experience during the 1990s stock market boom and bust cycle.

The subsequent sections of the paper are structured as follows. Section 2 provides an overview of the data utilized in our analysis and presents several facts about U.S. public pension investment behavior that serve as the foundation for our study. Section 2.3 introduces a portfolio choice model to frame potential drivers of these patterns, most notably the rise in the alternative-to-risky share. Section 3 focuses on the role of beliefs in the cross-section, while Section 4 considers risk-seeking channels. Section 5 evaluates the extent to which these mechanisms could account for the aggregate trend. Section 6 discusses agency-based and supply-side explanations, then Section 7 concludes. Additional details and results are available in an internet appendix.

2 Motivating Facts and Conceptual Framework

In this section, we begin by outlining the primary data sources used for our analysis. We then document that the aggregate alternative-to-risky share has risen sharply since the early 2000s, as has dispersion in the cross-section of U.S. public pensions. Finally, we analyze a standard mean-variance portfolio choice model–

[3] Few studies use CMAs from consultants and asset managers. Notable exceptions include Couts et al. (2023), who analyze the factor structure of expected returns, and Dahlquist and Ibert (2024), who study their cyclicality and impact on portfolio choice.

one commonly used by public pensions–to develop hypotheses about the potential mechanisms driving the rise of alternatives.

2.1 Sample description and variable construction

2.1.1 U.S. Public Pensions

Our primary data source is the January 2023 release of the Public Plans Database (PPD), which is maintained by the Center for Retirement Research (CRR) at Boston College. These data are based on comprehensive annual financial reports (CAFRs) that are filed annually by each defined-benefit (DB) public pension in the United States. The exact content and format of CAFRs vary across pensions and years, but all contain data on various plan characteristics like assets under management, portfolio composition, expected asset returns, actuarial value of liabilities, contribution rates, and information on beneficiaries. Throughout this paper, we refer to the expected rate of return on pension assets as an "asset hurdle rate" or "liability discount rate" because pension liabilities are discounted using the expected rate of return on pension assets under GASB 25 accounting standards.

Data in the PPD are reported at the plan level, though in many cases the assets of multiple plans are pooled and managed by pension "systems". For example, the board of the Colorado Pension Public Employees' Retirement Association invests and manages the pension assets of Colorado state employees, local school districts, the state's judicial system, and many local municipalities. Because our focus is on asset allocation decisions, our main unit of analysis is therefore at the system level. We map individual plans to larger pension systems based on hand-collected information and data from the Center for Retirement Research, and then aggregate plans to the system level accordingly.

Target Asset Allocation As a general rule, we analyze target portfolio weights whenever possible to focus on active portfolio choices and ensure consistency with the model in Section 2.3. Public pensions typically establish target portfolio shares through comprehensive asset-liability studies and mean-variance analysis (Campbell, 2018). These target portfolios—often referred to as policy portfolios—are then used as an input to performance evaluation and benchmark construction, liability accounting (Novy-Marx and Rauh, 2011), and rebalancing policy. When target shares are missing, we substitute them with actual shares. This approach is justified by the fact that public pensions, like other institutional investors, typically follow

rebalancing policies that are triggered when actual shares drift too far from target (NASRA, 2020).[4] With that said, such instances are uncommon in the PPD, as target shares are available for 89% of pensions by count and 94% by assets. For completeness, in Internet Appendix C, we also show that our main findings are robust when imputing missing target shares based on the approach in Calvet, Campbell, and Sodini (2009).

Target portfolio shares in the PPD are available for cash, fixed income, public equities, private equity (PE), real estate (RE), hedge funds (HFs), commodities, miscellaneous alternatives, and an asset class labeled "other." We combine real estate and commodities into a single asset class called real assets. For most of our analysis, we further aggregate these asset classes into three broad categories: (i) fixed income, which encompasses cash; (ii) public equities; and (iii) alternatives, comprising private equity, real assets, hedge funds, miscellaneous alternatives, and the "other" category. The risky share is defined as the sum of public equities and alternatives. Internet Appendix A.1 contains more detail on how we clean and process asset allocation data in the PPD, including adjustments we make to ensure asset class definitions are harmonized across time and pensions.

Consultants Most public pension investment boards rely on an outside consultant to provide advice on portfolio construction. The nature of advice provided by consultants can range from broad asset allocation decisions to specific portfolio manager selection (Andonov et al., 2023). Our focus in this paper is on broader asset allocation decisions. For this reason, we hand-collect data on the identity of the general consultant used by each pension for each year in the PPD data. In some cases, this information can be found easily on publicly available comprehensive annual reports. When it is not available from publicly available sources, we file Freedom of Information Act requests (FOIAs) with the pensions directly, asking for the "identity of the consultant that primarily advises on broad asset allocation decisions (e.g., percent in public equities, fixed income, private equity, etc.)." These collection efforts produce a fairly high coverage rate in our sample, as we have general consultant information for over 98% of the system-year observations in the PPD data. After accounting for mergers and name changes, there are 57 distinct consultants that have multiple public pension clients in a given year.[5]

After matching consultants with pension systems, we extract consultant beliefs using annual reports on Capital Market Assumptions (CMAs) that are published by most major consultancies. A typical CMA

[4] Internet Appendix A.1 contains a deeper discussion of the rebalancing policies of institutional investors.

[5] We assume that no consultant was used when we could not find information for a pension in a given year.

states beliefs about expected returns, volatilities, and correlations of different asset classes (Couts et al., 2023). Depending on the consultant, these beliefs capture risk-return properties for the next five to thirty years. The 15 consultants for whom we have CMAs manage the large majority of U.S. pension assets (~78%) but are more limited in total count. Additional details on the CMA data and how we process it can be found in Internet Appendix A.3.

Additional Variables Some of our subsequent empirical analysis relies on information about the board composition and Chief Investment Officers (CIOs) of U.S. public pensions. Board composition is based on Andonov et al. (2018) and downloaded directly from Aleksandar Andonov's website.[6] CIO identities are taken from Lu et al. (2023)

Data from the 1990s Our analysis in Section 3.2 draws on data from the 1990s for public pensions, sourced from the PENDAT surveys that were administered by the Public Pension Coordinating Council (see, for instance, Mitchell et al., 2001). These surveys, conducted in several waves during the 1990s, cover virtually all U.S. pensions by assets. They embed a wide range of information about each pension system, including cash flow and balance sheet data, portfolio construction, investment restrictions, and governance (Zorn, 1997; Mitchell et al., 2001). For the purposes of our analysis, we aggregate the plan-level data to the state level to reduce the influence of smaller pensions, which tend to have more missing data. Internet Appendix A.4 provides a detailed account of the structure and processing of these data. When target shares are missing in the PENDAT data, we first attempt to obtain them through FOIA requests submitted to the relevant pension systems. If these efforts are unsuccessful, we follow the same approach as with the PPD and substitute missing target shares with actual shares.

Summary Statistics Table 1 presents summary statistics for our sample of U.S. public pension funds in the PPD, broken out over four equally-spaced time periods. We discuss these summary statistics and our processing of the PPD data in greater detail in Internet Appendix A.1, where we also validate it against data from the U.S. Census Bureau. From 2006 onwards, the PPD covers over 90% of total U.S. public pension assets. The table further shows that PPD coverage is also fairly large in comparison to the broader U.S. pension system (roughly one-quarter), defined as the total amount of assets held by all private and public

[6]The Nebraska Investment Council enters the Andonov et al. (2018) data in 2005. FOIA responses confirm that its board structure was stable from 2001 to 2005, so we backfill earlier years using the 2005 data.

sector pension funds in Table L.117 of the Financial Accounts of the United States.

2.1.2 Other Institutions

Section 5.3 analyzes aggregate portfolio trends for U.S. corporate DB pensions, U.S university endowments, and U.K. corporate DB pensions. Data for U.S. corporate pensions are based on the corporate pension funding study by Milliman, which contains asset allocation data for the top 100 U.S. corporate DB pensions by assets. These pensions held $1.8 trillion of assets as of 2021. Asset allocation data for U.K. corporate DBs come from the U.K. Pension Protection Authority and U.S. endowment information comes from the National Association of College and University Business Officers (NACUBO). For each of these institutions, we use actual instead of target portfolio shares due to data limitations.[7] Internet Appendix A contains more background on these sources and their aggregate coverage.

2.2 Trends in Portfolio Composition

We now present some basic facts about the investment behavior of U.S. public pensions, starting with the overall risky share and then turning to the composition of risky investments.

2.2.1 The Risky Share in the Aggregate and Cross-Section

We define the risky share as risky investments over total AUM risky. Risky investments are defined as all holdings outside of cash and fixed income. This is an imperfect measure of the risky share because high-yield debt securities should be included in risky investments, yet granular data on the credit rating of fixed income investments are limited (see Internet Appendix A.1.3). Consistent with prior work on public pensions (e.g., Andonov, Bauer, and Cremers, 2017), Figure 1a shows that the target risky share has risen steadily since the turn of the century, albeit at a relatively modest pace. From 2001 to 2021, it went from 69% to 76%.

Figure 1b provides a longer-term perspective using the U.S. Census Bureau's Quarterly Survey of Public Pensions (QSPP), which covers the largest 100 pensions in the country starting in 1968. The first thing to notice is that risky shares from the QSPP and PPD largely overlap when both are available. Any differences between the two are likely attributable to the fact that target shares are plotted from the PPD, whereas the

[7] The distinction between actual and target shares matters less for long-run trends, as institutional investors typically rebalance toward targets after major market moves (see Internet Appendix A.7).

QSPP is based on actual shares. Second, and more strikingly, the 8 pp increase in the risky share since the 2000s is small compared to the 35 pp increase that occurred between 1970 and 2000. The risky share increased by nearly 17 pp in the 1990s alone, most of which was driven by a rotation out of U.S. Treasuries and into public equities (see Internet Appendix B.1.1). Figure 1b also suggests that the risky share for U.S. public pensions may have reached a new steady state of just under 80% in the last five years. We explore the cross-section of the risky share in Internet Appendix B.1.2.

2.2.2 The Alternative-to-Risky Share in Aggregate

While the increase in the risky share since the 2000s has been relatively modest, there has been a considerable change in the composition of risky investments. To illustrate this more clearly, Figure 2a starts by plotting the raw target portfolio weights for fixed income, public equities, and alternatives based on PPD data. The first thing that stands out from the plot is the rise of alternatives. From 2001 to 2021, the share of alternatives in the national portfolio increased from 9% to 30%, mirroring a broader trend by pensions around the world (Ivashina and Lerner, 2018; Betermier et al., 2021). At the same time, the target share of public equities fell from 59% to 46%. These flows imply that for every dollar that has shifted out of fixed income since 2001, $2.72 has moved into alternatives and $1.72 has flown *out* of public equities.

The preceding decomposition may overstate how much pension capital has moved out of public equities if hedge funds ultimately invest in the stock market. Figure 2b sheds some light on this issue by breaking out alternatives into subcategories for both 2001 and 2021. Hedge fund exposure has indeed increased by 5.4 pp during this period, though this change is not large enough to offset the contemporaneous decline of 13.1 pp in the public equity share. Thus, even if hedge funds are fully invested in public equities, public pensions have still decreased their target exposure to the stock market by 7.7 pp.

Figure 2b further shows that all forms of alternatives have risen since the turn of the century. In 2001, the respective target shares of real assets, private equity and credit, and hedge funds were 4%, 4%, and 0%. In 2021, their respective shares were 12%, 10%, and 6%.

Figure 2c plots the evolution of the target alternative-to-risky share (as opposed to raw shares). This object is a useful summary statistic for our purposes because it depends solely on beliefs in models where the two-fund separation theorem of Tobin (1958) holds (see Section 2.3). From 2001 to 2021, the target alternative-to-risky share rose from 14% to 39%.

2.2.3 The Alternative-to-Risky Share in the Cross Section

We now analyze how the composition of risky investments varies across public pensions, which we summarize using the target alternative-to-risky share $\Omega_{p,t}$ for pension p in year t. Figure 3a shows the distribution of $\Omega_{p,t}$ for each even year since 2001. There is a sizable amount of cross-sectional variation in $\Omega_{p,t}$: in 2021, the alternative-to-risky shares for the 10th and 90th percentile pensions were 18% and 57%, respectively. This dispersion has also widened considerably over time. In 2001, the spread between the 10th and 90th percentile pensions was 26 pp.

Figure 3b shows the distribution of changes in $\Omega_{p,t}$ across pensions systems from 2002 to 2021. Echoing the widening dispersion in Figure 3a, the shift into alternatives has varied strongly across pensions: the 25th percentile pension increased its alternative-to-risky share by 15 pp whereas the 75th percentile pension's share increased by 35 pp. Table A1b of the Internet Appendix shows that these changes have resulted in some amount of turnover in terms of the pensions with the highest alternative-to-risky share. For example, 21% of the pensions who were in the top quartile of $\Omega_{p,t}$ in 2021 were in the bottom quartile in 2002. At the same time, 13% of pensions who were in the bottom quartile of $\Omega_{p,t}$ in 2021 were also in the top quartile in 2002.

In Internet Appendix Section B.2, we discuss specific outlier pensions in terms of alternative usage and repeat our cross-sectional analysis on the portfolio share of alternatives (as opposed to the alternative-to-risky share).

2.2.4 Heterogeneity in Different Types of Alternatives

Pensions differ not only in how much they allocate to alternatives overall but also in which types of alternatives they invest. Figure 3c illustrates this heterogeneity in 2021, the final year of our sample. Each box plot depicts the distribution of the target share of a specific alternative subcategory relative to the total target share of alternatives. For example, the blue box plot shows the cross-sectional variation in the target PE-to-alternative share. At the 25th percentile, pensions allocate approximately 20% of their alternative portfolio to private equity, whereas at the 75th percentile, this number is closer to 40%. A similarly high degree of heterogeneity is observed for real assets (red box plot) and hedge funds (green box plot).

2.3 Conceptual Framework

The previous subsection documented a sharp rise in the level and dispersion of the alternative-to-risky share. We now look to theory for guidance on the potential factors driving these patterns. Our hypothesis development is grounded in standard mean-variance theory for a very simple reason: out of the top 100 largest pensions by size, 85 mention mean-variance analysis in their CAFRs, asset-liability studies, or documents available on their websites.[8]

We specifically model a two-period, power-utility investor in the tradition of Markowitz (1952) and Merton (1969). The investable universe consists of three assets: a riskless asset, public equities, and alternatives. The log return on each is denoted by r_f, r_E, and r_A, respectively. Public equity and alternative returns are jointly log-normal. Let μ denote the mean vector and Σ the variance-covariance matrix of log excess returns, with alternatives indexed first.

In what follows, it is useful to express the elements of μ and Σ as parameters from a CAPM-style regression of alternatives on public equities: $r_A - r_f = \alpha + \beta(r_E - r_f) + \eta_A$, where η_A is the idiosyncratic return to alternatives that has a volatility of σ_η. This implies $\mu_A = \alpha + \beta\mu_E$, $\sigma_A^2 = \beta^2\sigma_E^2 + \sigma_\eta^2$, and $\sigma_{AE} = \beta\sigma_E^2$. In Internet Appendix D.1, we extend the model to incorporate Bayesian learning about α from consultants, following an approach similar to Anagol et al. (2021).

The investor chooses portfolio weights $\omega = [\omega_f, \omega_A, \omega_E]'$ in each asset class to maximize the expected utility of terminal wealth. As shown in Campbell and Viceira (2002), the optimal portfolio is equivalent to that of a myopic, long-lived investor who ignores intertemporal hedging motives. The solution implies the following optimal weights on risky assets:

$$\begin{bmatrix} \omega_A \\ \omega_E \end{bmatrix} = \frac{\Sigma^{-1}\tilde{\mu}}{\gamma}, \tag{1}$$

where γ is the coefficient of relative risk aversion and $\tilde{\mu} = \mu + [\sigma_A^2/2, \sigma_E^2/2]'$ accounts for Jensen's Inequality. We define the alternative-to-risky share as:

$$\Omega := \frac{\omega_A}{\omega_A + \omega_E} = \frac{\Sigma^{-1}\tilde{\mu}e}{\Sigma^{-1}\tilde{\mu}o}, \tag{2}$$

[8] To determine usage, we searched across websites and public filings of pensions from 2016 to 2024. We also searched for pension materials on the eVestment database.

where $e = [1,0]'$ and $o = [1,1]'$. Complete derivations and details of the model are provided in Internet Appendix I.2–I.3.

Next, we explore two possible mechanisms within this model that could explain the rise of alternatives (i.e., an increase in Ω).

Beliefs The first mechanism is a shift in the perceived alpha α of alternatives relative to public equities. In our model, the alternative-to-risky share Ω is increasing in α:

$$\frac{\partial \Omega}{\partial \alpha} = \frac{(\beta \omega_A + \omega_E)}{\gamma \sigma_\eta^2 (\omega_A + \omega_E)^2} > 0, \tag{3}$$

so long as $\beta \omega_A + \omega_E > 0$, a condition that is trivially satisfied in practice. Intuitively, when the investor perceives a higher alpha for alternatives, it shifts the composition of its risky portfolio toward that asset class independent of risk aversion γ.[9] This result also means that cross-sectional dispersion in the alternative-to-risky share is modulated by the degree of disagreement about alpha.

Constrained Risk-Taking The second mechanism is a decline in risk aversion, γ, or an increased desire to take risk. In practice, such a shift could reflect factors outside the model, such as underfunding, reach-for-yield triggered by low interest rates, or spending constraints (Campbell and Sigalov, 2022). Importantly, Equation (2) makes clear that in the unconstrained model, the optimal alternative-to-risky share Ω is *independent* of risk aversion. This is the essence of Tobin's separation theorem: as risk aversion falls, the investor scales up exposure to the risky portfolio but does not alter its composition. Thus, in the unconstrained model, the rise of alternatives cannot be explained by increased risk tolerance alone.

This separation result breaks down if there is a constraint on the maximum risky share, $\omega_A + \omega_E \leq \omega_{max}$. For the remainder of the paper, we refer to this as a leverage constraint, even if $\omega_{max} \leq 1$. When the constraint binds, the alternative-to-risky share becomes sensitive to risk aversion:

$$\frac{\partial \Omega}{\partial \gamma} = -\frac{K}{\omega_{max} \gamma^2},$$

where K is a function of beliefs (see Internet Appendix I.3 for more details). For some initial beliefs ($K > 0$), a more risk-tolerant investor may increase its alternative-to-risky share if it cannot increase its risky share.

[9] Though risk aversion γ appears in Equation (3), it is straightforward to show that it actually cancels out because Equation (1) shows that ω_A and ω_E are both functions of γ.

In particular, this will happen if alternatives offer higher expected returns than public equities but do not contribute too heavily to portfolio variance on the margin.[10]

In summary, the textbook mean-variance model used by most public pensions suggests two potential explanations for the rise of alternatives. The first is that pensions have become more optimistic about the alpha of alternatives relative to public equities. The second is that pensions have become more risk-seeking but are constrained from increasing their risky share. In what follows, we look for empirical markers of both explanations, though of course these channels are not mutually exclusive.

3 Belief-Based Explanations

In this section, we evaluate the belief-based mechanism. We first show that the alternative-to-risky share tends to be higher for pensions whose investment consultants believe alternatives have a high alpha. An instrumental variables design further reveals that this positive correlation is largely due to the causal impact that consultants have on their clients. Our estimates suggest that widening disagreement among consultants about alpha accounts for about a third of the increased cross-sectional dispersion in the alternative-to-risky share since the early 2000s. We then examine two classic factors that influence beliefs: past experiences (Malmendier and Nagel, 2016) and peer behavior (Bailey et al., 2022).

3.1 Evidence from Pension Consultants

3.1.1 The Time-Series of Consultant Beliefs

The analysis in this subsection relies heavily on beliefs held by pension consultants about the alpha of alternatives relative to public equities. Figure 4a depicts how the average perceived alpha $\bar{\alpha}_t$ has evolved over time. For each year t, we first compute consultant-average alphas for private equity ($\bar{\alpha}_{pe,t}$), real estate ($\bar{\alpha}_{re,t}$), and hedge funds ($\bar{\alpha}_{hf,t}$).[11] $\bar{\alpha}_t = (\bar{\alpha}_{pe,t} + \bar{\alpha}_{re,t} + \bar{\alpha}_{hf,t})/3$ is then defined as their equal-weighted average. This construction ensures that variation in $\bar{\alpha}_t$ is driven entirely by variation in the perceived alpha of different asset classes, rather than shifts in the composition of alternative assets. To maintain a relatively

[10] In Section 5, we calibrate the model to the aggregate portfolio in 2001 and then simulate a decline in risk aversion coupled with a binding constraint on the risky share. The key finding is that, absent a shift in beliefs, a more risk-seeking investor would reallocate toward public equities and away from alternatives. The reason is that the low share of alternatives in 2001 implies that pensions initially viewed alternatives less favorably than public equities (i.e., $K < 0$).

[11] The alpha $\alpha_{a,c,t}$ of consultant c for asset class a in year t is constructed by plugging in c's CMA data on expected returns and covariances into the usual CAPM formula (see Internet Appendix A.3). We limit our analysis to these three asset classes because CMA data on infrastructure and private credit are only available in the last few years of the dataset.

consistent panel, we include only consultants with at least ten years of data. The time series begins in 2003, the first year in which beliefs about all three asset classes are available for multiple consultants.

Figure 4a shows that consultants have become more bullish on the alpha of alternatives over time, with $\bar{\alpha}_t$ rising by 58 bps from 2003 to 2021. In Internet Appendix D.2, we show formally how to decompose this change into three main sources: (i) an increase in average expected excess returns of alternatives; (ii) a decline in the average public equity risk premium, defined as the expected excess return of public equities; and (iii) a decline in the beta of alternatives with respect to public equities—equivalently, an improvement in the perceived diversification properties of alternatives.[12]

Table 2 provides a high-level sense of how much each of these channels has contributed to the upward trend in Figure 4a. The top row of the table focuses on the (equal-weighted) average alpha $\bar{\alpha}_t$ across all types of alternatives. The summary statistics reveal a clear pattern: the increase in $\bar{\alpha}_t$ has been almost entirely driven by an increase in expected returns, as the two have moved in tandem since 2003. At the same time, the average perceived beta $\bar{\beta}_t$ of alternatives has increased slightly to just over 0.6. This increase suggests that an improvement in the perceived diversification benefits of alternatives cannot explain the increase in $\bar{\alpha}_t$. If anything, perceived diversification benefits have declined over time.

3.1.2 Transmission to Pension Clients

Next, we study how consultant beliefs about alpha shape the portfolio choices of their clients. Estimating this casual effect is valuable for at least two reasons. First, it allows us to directly measure how much of the increased cross-sectional dispersion in the alternative-to-risky share is attributable to widening disagreement among consultants about alpha. Second, as we discuss later in Section 5.2, it sheds light on the extent to which consultant beliefs have contributed to the aggregate rise in the alternative-to-risky share.

In Internet Appendix D.1, we microfound the link between consultant beliefs and client outcomes by incorporating Bayesian learning into the mean-variance model from Section 2.3. In the extended model, each pension p begins with a prior belief about alpha and updates that belief after receiving a noisy signal $\alpha_{c(p)}$ from its consultant $c = c(p)$. We then show that the alternative-to-risky share can be linearly approximated as follows:

$$\Omega_p \approx b_0 + \underbrace{b_1 \alpha_{c(p)}}_{\text{Causal effect of consultants}} + \underbrace{\mathbf{b}_2' \Psi_p}_{\text{Prior Beliefs}} + \underbrace{b_3 v_p}_{\text{Non-belief motives}}, \tag{4}$$

[12] Technically, a fourth residual source reflects changes in the cross-consultant covariance between beta and the public-equity risk premium. See Internet Appendix D.2 for details.

where Ψ_p is a vector that parameterizes each pension's prior beliefs about alpha (e.g., its prior mean), and ν_p is an idiosyncratic preference for alternatives that is orthogonal to beliefs (e.g., agency frictions). This non-belief preference ν_p can be thought of as arising from outside the Bayesian learning model or as implicitly affecting the pension's risk aversion. $b_0, b_1, \mathbf{b}_2$, and b_3 are reduced-form objects determined by the model's structural parameters.

Next, suppose we estimate b_1 using an OLS regression of Ω_p on consultant beliefs $\alpha_{c(p)}$. In this case, the population regression coefficient β is given by the standard omitted variable bias formula:

$$\beta = b_1 + \tau_B + \tau_{NB},$$

where $\tau_B = \mathbb{C}(\mathbf{b}_2'\Psi_p, \alpha_{c(p)})/\mathbb{V}(\alpha_{c(p)})$ captures bias due to selection on prior beliefs, and $\tau_{NB} = \mathbb{C}(b_3\nu_p, \alpha_{c(p)})/\mathbb{V}(\alpha_{c(p)})$ captures bias due to selection on non-belief factors. The former bias could arise if pensions choose consultants who share their beliefs, while the latter could occur if pensions seek to justify agency-driven preferences by hiring consultants who are bullish on alternatives. Both forms of selection will cause the OLS coefficient β to diverge from the causal parameter b_1. Thus, for the purposes of estimating the causal impact of consultant beliefs on the alternative-to-risky share, we need an identification strategy to handle this potential omitted variable bias.

With these identification issues in mind, we estimate the following panel analogue of Equation (4):

$$\Omega_{p,t} = \lambda_t + \beta\alpha_{c(p),t} + \Gamma X_{p,t} + \varepsilon_{p,t}, \tag{5}$$

where $\Omega_{p,t}$ is the target alternative-to-risky share of pension p in year t and $\alpha_{c(p),t}$ is the equal-weighted average alpha of PE, RE, and HFs reported by pension p's consultant c in year t. If beliefs about a specific type of alternative are missing, $\alpha_{c(p),t}$ is constructed using the equal-weighted average alpha of the available types. Because it does not vary across pensions that share a consultant in a given year, there is no mechanical relationship between $\alpha_{c(p),t}$ and $\Omega_{p,t}$ in regression (5).

In regression (5), λ_t is a time fixed effect and $X_{p,t}$ is a vector of controls that includes the level of pension funding, (log) size, the expected return on each pension's assets (i.e., its hurdle rate), the ratio of required actuarial contributions to payroll, the ratio of total deductions to payroll, the cash share, and the annual return of the fund. $X_{p,t}$ also includes the difference between each pension's target and actual risky share,

residualized to its past return. These controls are designed to capture agency-based and risk-seeking motives that may lead some pensions to prefer alternatives over public equities (v_p). Their logic is discussed further in Sections 4.1 and 4.2. By including them, we seek to minimize the bias on β induced by selection on non-beliefs. Standard errors are double-clustered by consultant and time.

Column (1) of Table 3 presents OLS estimates of Equation (5) when including no controls. The point estimate of 3.39 ($t = 2.98$) is statistically significant and indicates that pensions have higher alternative-to-risky shares when their consultants believe the alpha of alternatives to be high. Figure 4b depicts this relationship using a binscatter plot and confirms our linear specification.

Column (2) shows OLS estimates when adding controls $X_{p,t}$ to the regression. The point estimate of 3.45 is very similar to column (1) and remains statistically significant ($t = 3.19$). Because the controls are designed to capture non-belief preferences for alternatives, the fact that the coefficient remains stable when moving from column (1) to (2) suggests that selection on non-beliefs is likely to be small. We test this idea more formally by applying the bias-adjustment from Oster (2019), under her recommended assumption that the controls $X_{p,t}$ and any unobserved variables related to non-belief preferences contain equal explanatory power for $\alpha_{c(p),t}$. The resulting bias-adjusted coefficient on $\alpha_{c(p),t}$ in this case still equals 3.49. The economic significance of this point estimate is discussed below.

While the controls in column (2) mitigate selection on non-beliefs, they are unlikely to address selection on beliefs. To tackle this issue, we build an instrument for $\alpha_{c(p),t}$ based on the findings of Andonov et al. (2023), who show that pensions are more likely to hire geographically proximate consultants. In our setting, we define proximity based on whether a consultant c is active in pension p's state $s(p)$. A consultant is considered active in state s if it employs at least one "Investment Advisory Representatives" (IAR) who is either registered to operate or maintains their home office in that state. Information on the registration and employment status of IARs is derived from an updated version of the dataset used by Egan et al. (2019).[13]

Formally, we define the instrument for each consultant-pension pair as follows:

$$z_{c(p),k,t} = \sum_{j=1}^{C} w_{j,p,k} \alpha_{j,t},$$

where j indexes consultants, $\alpha_{j,t}$ is the belief of consultant j in year t, and $w_{j,p,k}$ is a weight based on whether

[13] We thank Mark Egan for sharing these data with us. We further assume that a consultant is active in a state if it has a pension client in that state based on CAFRs. Complete details on the instrument are given in Internet Appendix Section D.3.1, including a discussion of the costs of becoming an IAR.

consultant j is active in p's state in year k. More specifically, define $N_{p,k}$ as the number of consultants that are active in pension p's state as of time k. For pension p in year t, we set $w_{j,p,k} = N_{p,k}^{-1}$ for consultants that are active in p's state as of year k and $w_{j,p,k} = 0$ for those that are not. This ensures that $z_{p,k,t}$ is constructed by averaging over the active consultants in p's state. For reasons that will become clear shortly, our notation here explicitly distinguishes between the year in which the weights are defined, k, and the year in which beliefs are measured.

$z_{c(p),k,t}$ should be thought of as a Bartik instrument (Bartik, 1991; Blanchard and Katz, 1992), whose cross-sectional variation is driven entirely by the weights, $w_{j,p,k}$. Goldsmith-Pinkham et al. (2020) show that this instrument will satisfy the usual exogeneity condition if, conditional on controls $X_{p,t}$, the weights in year k—equivalently, the states in which consultants are active—are exogenous to pension preferences for alternatives, whether belief-driven ($\Psi_{p,t}$) or not ($\nu_{p,t}$).

Given this condition, we construct weights in two different ways. In the first, we set $k = t$, so that $z_{c(p),k,t}$ in year t is defined using weights that are based on the contemporaneous activity of each consultant. The logic of this weighting scheme is that a consultant's state co-location in a given year t is exogenous to contemporaneous pension preferences for alternatives, *after* conditioning on $X_{p,t}$. One may object that consultants have incentives to be geographically close to large pensions given their fees are typically based on AUM, but we directly control for size in $X_{p,t}$. The second weighting scheme sets $k = 2005$, the year before the shift to alternatives accelerated (Figure 2c). The instrument satisfies the relevance condition if pensions are more likely to select consultants who are active in their state.

Column (3) of Table 3 presents instrumental variable (IV) regression estimates of Equation (5) when instrumenting $\alpha_{c(p),t}$ with the $z_{c(p),s,t}$ that is based on contemporaneous distances ($s = t$). The large first-stage F-statistic confirms the relevance of our instrument and suggests that pensions are more likely to hire nearby consultants, consistent with Andonov et al. (2023). The estimated β_{IV} equals 3.27 ($t = 4.24$) and is statistically significant. The third row from the bottom of the table shows that the IV estimate is slightly lower than the OLS estimate, indicating that selection is positive yet small on average.

Column (4) shows IV estimates using the instrument based on where consultants were active in 2005. This instrument is more likely to satisfy the exogeneity condition because it is unlikely that consultants selected their geographic footprint in 2005 based on *future* pension preferences for alternatives. The point estimate here equals 4.01 and remains statistically significant ($t = 5.27$). To contextualize this magnitude, note that the spread between the 90th and 10th percentile consultant-reported α expanded by roughly 100

basis points from 2001 to 2021. Accordingly, the point estimate in column (4) suggests this widening contributed to a 4 pp increase in the cross-sectional dispersion of the alternative-to-risky share. During the same period, the gap in the alternative-to-risky share between the 10th and 90th percentile pension grew by about 13 pp. Therefore, assuming our IV is valid, about one-third of the variation in adoption rates of alternatives across pensions can be attributed to widening disagreement among consultants regarding the alpha of alternatives.

One concern with the standard errors in Table 3 is that they are based on a relatively small number of consultant clusters (Cameron et al., 2008). To address this potential issue, we apply the wild bootstrap procedure proposed by Cameron et al. (2008) to test the null hypothesis that the estimated β is equal to zero. The resulting p-value is reported in the row labeled $p_{wild}(\beta = 0)$ for all specifications in the table and indicates that we can reject the null at conventional confidence levels. As a complementary approach, Cameron and Miller (2015) recommend basing inference on critical values drawn from the Student's t-distribution with $G-1$ degrees of freedom, where G is the number of clusters. For $G = 15$, the critical t-value at a 0.05 significance level is 2.145—well below the lowest t-value 2.98 reported in Table 3. Even with as few as $G = 5$ clusters, a t-statistic of 2.98 would still suffice to reject the null hypothesis of $\beta = 0$ at a 5% significance level.

Section D.1.5 of the Internet Appendix uses the Bayesian model extension to explore how idiosyncratic pension skill—defined following Anagol et al. (2021) as the relative precision of the consultant's signal—modulates the causal influence of consultants. In our context, skill can be thought of as an ability to independently and evaluate alternative investments. Intuitively, when a pension is relatively uncertain about its own prior (low-skill), the consultant's influence on the raw alternative share becomes stronger.[14] Empirically, we proxy for skill using overall size and board composition (Andonov et al., 2018). Consistent with the model, we find that smaller pensions with more politicized boards are nearly twice as responsive to their consultant's beliefs compared to larger pensions with less political boards.

3.1.3 A Variance Decomposition

To complement our IV design, we now use a series of fixed-effects regressions to assess how much of the cross-sectional dispersion in the alternative-to-risky share is attributable to disagreement about alpha. This

[14]The effect on the alternative-to-risky share is more theoretically ambiguous because a low-skilled pension that updates positively about alpha after meeting its consultant will increase its raw alternative and total risky shares.

analysis is agnostic about the source of disagreement, which could be driven by consultants, pensions, or both. Our workhorse panel specification mirrors Equation (5) but replaces consultant-reported beliefs with consultant fixed effects, λ_c:

$$\Omega_{p,t} = \lambda_t + \lambda_c + \Gamma X_{p,t} + \varepsilon_{p,t}. \tag{6}$$

The advantage of this approach is that it can be estimated for our full sample of pensions matched with consultants, whereas CMA data are available for 63% of system-year observations.

The objects of interest in the regression are the consultant fixed effects λ_c. As we show in Internet Appendix D.4, the incremental R^2 from including λ_c provides a lower bound on the cross-sectional variation in the alternative-to-risky share that is attributable to beliefs. This bound holds under the assumption that, after conditioning on observables $X_{p,t}$, pensions do not select into consultants on basis of non-beliefs (e.g., risk-seeking motives). Intuitively, absent selection on non-beliefs, variation in λ_c can only reflect differences in consultant beliefs, pension beliefs, or selection based on beliefs. We provide evidence in support of this assumption in Internet Appendix D.4.3, where we study the portfolio composition of public- and private-sector clients who share a consultant.[15]

Table 4 summarizes the results. Adding consultant fixed effects to the regression increases the adjusted R^2 by 17 pp, implying that beliefs explain at least this much of the cross-sectional variation in the alternative-to-risky share. The true share will be higher if there is meaningful belief heterogeneity across pensions that share a consultant (see Internet Appendix D.4). The F-statistic and the p-value in the table indicate that we can reject the null hypothesis of equal consultant effects with high statistical confidence. Figure 5 visualizes the wide distribution of the consultant fixed effects, after applying Bayesian shrinkage to account for sampling error (Casella (1992), Eqs. 7.11 and 7.13). Clients of the 5th percentile consultant have an average alternative-to-risky share of 7%, whereas clients of the 95th percentile consultant have an average share of 50%.

One concern is that $X_{p,t}$ may not adequately control for non-belief motives that could drive pension preferences for alternatives. An obvious example is that some state legislatures allowed for local pensions to invest in alternatives later than others (e.g., South Carolina). A natural way to handle state-level confounders of this kind is to replace time fixed effects in Equation (6) with state-by-time fixed effects. Even under this

[15] The basic idea is that public- and private-sector pensions face different agency frictions because they have different governance structures, accounting rules, and regulations. Thus, if public- and private-sector clients who share a consultant invest similarly, it is more likely driven by beliefs rather than selection based on agency frictions.

more stringent specification, consultant effects add 12 pp of incremental R^2 to the regression model, further confirming that beliefs (via consultants) are a key driver of alternative use across pensions.

Rows (5) to (16) of Table 4 repeat this exercise by running regression (6) for different types of alternatives, not just the alternative-to-risky share. Rows (6), (10), and (14) show there are also large consultant effects for private equity, hedge fund, and real asset investments. This conclusion is robust to adding state-by-time fixed effects. The incremental R^2 from adding consultant effects indicates that beliefs account for at least 14-19% of the cross-pension variation in the use of specific types of alternatives (see Internet Appendix E.2 for additional analysis).[16]

3.2 Experience

Our analysis has emphasized the role of beliefs—particularly those shaped by investment consultants—in explaining cross-pension differences in alternative investment intensity. A growing literature shows that beliefs are also shaped by experience. In the context of public pensions, Andonov and Rauh (2022) find that past performance influences expectations about portfolio returns. Building on this work, we examine how experience during the 1990s influences the *composition* of risky assets.

The 1990s were a formative period for U.S. pensions, many of which significantly increased their exposure to public equities for the first time (see Figure 1b).[17] Stock markets experienced a major boom-and-bust cycle, ending with the dot-com crash in 2000–2002. Pensions that shifted into equities later in the decade were more exposed to this downturn, while early movers benefited more from the boom. We hypothesize that this negative early experience made late entrants subsequently more bearish on public equities, leading them to increase their alternative-to-risky share more than those with more favorable prior exposure.

One way to test this hypothesis is to compare two pensions, A and B, that increased their equity share by the same amount between 1990 and the bursting of the dot-com bubble in 2002. The key difference is that pension A increased its share later in the bull market and was therefore more influenced by the bubble's burst. We implement this idea using the following state-level regression:

[16] The consultant fixed effects in Table 4 speak to average differences in alternative use across pensions. To study adoption rates, we can instead add consultant-by-time fixed effects, although these are difficult to separate from time fixed effects. For the alternative, PE, HF, and RA shares (relative to the risky share), consultant-by-time fixed effects add 10%, 10%, 9%, and 8% of incremental R^2 to the regressions after including controls.

[17] According to the Annual Survey of Public Pensions (ASPP), virtually every dollar that left fixed income during the 1990s went into public equities (see Internet Appendix B.1.1. In the cross-section, this period marked the first meaningful exposure to equities for many pensions, let alone a majority allocation. In 1990, the 5th, 10th, and 25th percentiles of target stock exposure were just 0%, 11%, and 30%, respectively. By 2001, those percentiles had increased sharply to 44%, 46%, and 54%.

$$\Delta\Omega_{s,2002\to2021} = c + \beta_1 \Delta\text{Equity}_{s,1996\to2002} + \beta_2 \Delta\text{Equity}_{s,1990\to2002} + \Gamma X_s + \varepsilon_s \tag{7}$$

where $\Delta\Omega_{s,2002\to2021}$ is the change in state s's target alternative-to-risky share from 2002 to 2021 and $\Delta\text{Equity}_{s,1996\to2002}$ is the change in s's target public-equity share between 1996 and 2002. $\Delta\text{Equity}_{s,1990\to2002}$ is the change in the target share of public equities between 1990 and 2002. As mentioned in Section 2.1, all data from the 1990s are sourced from the PENDAT surveys (see Internet Appendix A.4 for complete details). X_s is a vector of state-level controls that we also describe below. The object of interest in the regression is β_1. Because we control for the total change in equities between 1990 and 2002 ($\Delta\text{Equity}_{s,1990\to2002}$), β_1 isolates the timing of the change. Under our hypothesized mechanism, it should therefore be positive. Robust standard errors are used in all regressions.

As a baseline, column (1) of Table 5 summarizes the results of a univariate regression of $\Delta\Omega_{s,2002\to2021}$ and $\Delta\text{Equity}_{s,1996\to2002}$.[18] Consistent with our hypothesized mechanism, β_1 is positive and statistically significant. Its magnitude implies that a one-standard deviation increase in $\Delta\text{Equity}_{s,1996\to2002}$ is associated with a 6.57 pp increase in $\Delta\Omega_{s,2002\to2020}$. The standard deviation of the latter equals 16.64 pp. Moreover, the change in equity share between 1996 and 2002 alone explains 16% of the variation in state-level changes in the alternative-to-risky share between 2002 and 2021.

Column (2) adds $\Delta\text{Equity}_{s,1990\to2002}$ as an additional control to the regression. Its inclusion is important because it holds fixed the total increase in equities during the boom-and-bust cycle and therefore isolates the timing of entry into public equities. In this case, the coefficient on $\Delta\text{Equity}_{s,1996\to2002}$ increases in magnitude relative to column (1) and remains statistically significant at conventional confidence levels.

One limitation of the PENDAT data is that target shares in 1990 are available for only 32% of pensions by count and 57% by assets.[19] Because actual shares are used when target shares are not available, $\Delta\text{Equity}_{s,1990\to2002}$ in column (2) is based largely on 1990 actual shares. In column (2) of Table 5, the row labeled "1990 method" makes this approach to handling missing target shares explicit. To ensure that this decision does not drive our results, we infer active changes in public equities between 1990 and 2002 using the decomposition method outlined in Calvet et al. (2009). Full details of this procedure

[18] Due to missing data, not all states are included in regression (7). Specifically, PPD data is unavailable for New Jersey, Rhode Island, and Vermont in 2002. Additionally, PENDAT data is missing for DC in both 1990 and 1996 and for West Virginia in 1990.

[19] Coverage improves significantly in 1996, where target shares are available for 72% of pensions by count and 89% by assets. The elevated likelihood of missing target shares among smaller pensions is one reason why we prefer to run regression (7) at the state-level, as this reduces the influence of smaller pensions.

are contained in Internet Appendix C. We denote this active change by $A^e_{s,1990\to2002}$. Column (3) sets $\Delta\text{Equity}_{s,1990\to2002} = A^e_{s,1990\to2002}$ for states in which over 50% of the assets are missing target shares. This substitution is flagged as "Active Change" in the "1990 method" row of the table. Column (3) shows that using $A^e_{s,1990\to2002}$ only strengthens our results, as both the magnitude and precision of the estimated β_1 increase relative to column (2). Similar results are obtained when using $A^e_{s,1990\to2002}$ for all states, regardless of the availability of target shares, and when using different replacement thresholds. Thus, substituting missing target shares in 1990 with actual shares does not appear to confound our interpretation of Table 5.[20]

Column (4) introduces several controls to the regression, including changes in hurdle rates and in the fraction of retired members (Andonov et al., 2017), the level of BEA-adjusted funding in 2002, and the log of AUM in 2002, and the ratios of deductions and required contributions to payroll in 2002 (Giesecke and Rauh, 2022). All changes are measured between 2002 and 2021. These variables are designed to control for variation in risk-seeking motives, especially those potentially induced by poor stock-market performance during the 1990s.

Column (4) also includes controls that capture the legal environment and financial sophistication. One control is an indicator for whether state constitutional investment restrictions were in place as of 1998. This accounts for the possibility that states increasing their alternative-to-risky share most aggressively in the 2000s were simply those that relaxed legal limits later (e.g., South Carolina). To address the concern that some plans are serial trend-chasers—first rushing into public equities during the 1990s bull market and then into alternatives in the 2000s—we add two proxies for financial sophistication: (i) an indicator for whether the plan imposed minimum professional qualifications on investment decision-makers as of 1998; and (ii) the fraction of political board members, following Andonov et al. (2018). All three variables are aggregated to the state level using asset-weighted averages. The estimated β_1 increases when adding these controls to column (4) and remains statistically significant at conventional confidence levels.

Lastly, as a placebo test, we replace the change in the alternative-to-risky share from 2002 to 2021 with the change in the risky share as the dependent variable in regression (7). The estimated β_1 is about one-quarter the magnitude of the coefficient in column (3) and is not statistically significant. This placebo test reinforces our argument that experience in the stock market during the 1990s—namely the bursting of the dot-com bubble—influenced pension beliefs about alternatives versus public equities, rather than their

[20] Our results are also not confounded by differences in fiscal year timing across pensions. When running the regression in column (3) for the subset of states in which all pensions have mid-year fiscal year ends ($N = 29$, we estimate that $\beta_1 = 0.99$ ($t = 3.42$).

overall willingness to take risk.

3.3 Peer Effects

Finally, we examine whether peer behavior helps explain cross-sectional variation in alternative use. Motivated by recent work in household finance showing that beliefs and choices are shaped by social networks (Bailey et al., 2018, 2022), we define peer networks geographically. Specifically, for each pension p in year t, we construct the target alternative-to-risky share of its peers as follows:

$$n_{pt} \equiv \sum_{k \neq p} \delta_{pk} \times \Omega_{k,t},$$

with weight $\delta_{pk} = d_{pkt}^{-1} \times \left(\sum_{j \neq p} d_{p,j,t}^{-1} \right)^{-1}$. The target alternative-to-risky share for pension k at time t is $\Omega_{k,t}$. d_{pkt} is the distance (in kilometers) between the headquarters of pension p and pension k in year t. We measure distance using the 5-digit zip codes of each pension in our sample.[21] The weights δ_{pk} are therefore based on the inverse distance between pension systems.

We then measure the degree of peer effects using variants of the following panel regression:

$$\Omega_{p,t} = \lambda_{cdt} + \beta_z n_{pt} + \theta' \mathbf{X}_{pt} + \varepsilon_{pt}, \tag{8}$$

where λ_{cdt} is a consultant-by-time-by-census-division fixed effect and $\mathbf{X}_{pt}$ is a vector of pension observables that includes each pension's GASB 25 funding ratio, asset hurdle rate, (log) size, required actuarial contribution relative to payroll, and administrative expenses relative to payroll. The fixed effect λ_{cdt} is included to isolate peer effects from the impact of shared consultants (Section 3.1.3) and local unobserved shocks that may affect portfolio choice (Angrist (2014)). The inclusion of λ_{cdt} means that β_z is identified from variation within the same consultant, census division, and year. Standard errors in the regression are clustered by state and time.

Column (1) of Table 6 shows a sizable and statistically precise estimate of peer effects. On average, a 10 pp increase in the alternative-to-risky share of a pension's peers is associated with a 7.1 pp increase in its own alternative-to-risky share. As we show later in Section 4.2, this elasticity further suggests that peer behavior explains the composition of risky investments much better than a pension's own funding level,

[21] For pension systems located in the same zip code, we assume they are 1.6 kilometers (1 mile) apart.

asset hurdle rate, or other proxies for risk aversion.

One interpretation of this finding is that pensions learn about the alpha of alternatives from nearby peers, perhaps through more frequent interactions among investment staff and CIOs at conferences or industry events. Internet Appendix E.1 provides a more detailed discussion of this channel and presents anecdotal evidence consistent with this interpretation. A second interpretation is based on the herding model of Scharfstein and Stein (1990), whereby pensions herd with their nearby peers to avoid public backlash for contrarian behavior. While both channels are likely operant in the data, we try to isolate the first channel by estimating peer effects in subsets of the data where herding incentives are likely to be weaker.[22]

Column (2) proxies for herding incentives using CIO tenure, based on the assumption that more established CIOs face less pressure to conform. Specifically, we construct an indicator for whether a CIO has at least five years of tenure—roughly the median in the post-2005 sample—and interact it with peer allocations in the regression. Because our CIO data begin in 2001, we restrict the sample to years after 2005 to allow for meaningful tenure measurement. The interaction term is negative, as expected, but is small in magnitude and statistically indistinguishable from zero.

In column (3), we proxy for herding incentives with an indicator variable for whether a pension is well-funded, defined as being in the top quartile of GASB 25 funding over our full sample. In practice, this means we only include pensions whose GASB 25 funding is above 88%. The assumption in this test is that well-funded pensions are under less public scrutiny and therefore have less incentive to herd. The negative interaction term in column (3) is consistent with this idea, but nonetheless implies that well-funded pensions are responsive to their peers.

The last measure of herding incentives that we consider is an indicator variable based on whether a pension is in the top quartile of annual net-of-fee performance, defined as the ratio of net investment income to beginning-of-year assets as in Andonov and Rauh (2022). Similar to column (3), the idea is that high-performing pensions face less public scrutiny, thereby reducing their incentive to herd as in Scharfstein and Stein (1990). The negative and statistically significant interaction term in column (4) suggests that this may be the case, but much like our previous results, we continue to find an economically and statistically meaningful peer pass-through coefficient of 0.55 ($t = 2.61$) for this subgroup of pensions.[23]

[22] A third possibility is that consultants themselves are the source of peer networks (e.g., they sponsor client-only events). However, we provide evidence against this interpretation in Internet Appendix E.1, where we show that peer effects remain just as strong when n_{pt} excludes pensions that share a consultant with p.

[23] Differences in fiscal year end dates across pensions can complicate cross-sectional rankings of performance. For robustness, we have also estimated a variation of the regression in column (4) in which 1_{pt}^{HP} is defined based on whether a pension has above-

For robustness, column (5) replaces the contemporaneous alternative-to-risky share of peers with its lagged value. Though imperfect, this is one way to allay fears that the regression coefficients are driven by reverse causality. The point estimate in column (5) is similar in magnitude to that in column (1) and remains statistically significant.

Our preferred interpretation of Table 6 is that peers shape how pensions form beliefs about the risk-return profile of alternatives. As is common in studies of peer effects (Angrist, 2014)), it is difficult to fully distinguish this story from one in which nearby pensions are exposed to common non-belief-driven shocks (e.g., changes in member life expectancy). To further reinforce the belief-driven channel, column (6) measures peer behavior based on the average alternative-to-risky share of endowments, corporate pensions, and unions in each state. These institutions differ widely from public pensions in their funding structure, governance, and regulation, thereby reducing the likelihood they are exposed to common shocks.[24] Consistent with belief-driven peer effects, the estimated β remains positive and statistically significant, and is comparable to the elasticity for high-performing pensions found in column (5). The idea that public pensions form beliefs about alternatives based on the behavior of local private-sector institutions seems plausible given the relatively high historical performance of these institutions, most notably endowments (Lerner et al., 2007, 2008). Indeed, university endowments adopted alternatives starting in the 1990s, following the so-called Yale Endowment model pioneered by David Swensen, who was famously bullish on alternatives (Goetzmann and Oster, 2012).

4 Risk-Seeking Explanations

In the mean-variance model from Section 2.3, pensions may shift their risky investments toward alternatives if they want to take on more risk but are constrained from doing so. In this section, we ask whether this mechanism can explain cross-pension variation in the adoption of alternatives. Our analysis consists of three parts. First, we develop several measures of risk constraints and estimate their relationship with long-run changes in the alternative-to-risky share. Second, we propose a few related mechanisms through which effective pension risk aversion might decline or vary in the cross-section, including factors such as pension

median net-of-fee returns relative to other pensions whose fiscal year ends in the same month. We use medians instead of quartiles in this specification because the comparison group for performance ranking becomes smaller when conditioning on the same fiscal year end. Under this alternative definition of relative performance, the estimated belief-driven peer effect equals 0.63 ($t = 2.68$).

[24] State-level data for private-sector institutions is sourced from S&P's Money Market Directory, which we discuss in detail in Section A.5 of the Internet Appendix. To ensure sufficient variation in this peer measure, we replace the consultant-by-census-division-by-year fixed effect in regression (8) with a consultant-by-year fixed effect.

underfunding. We then connect various proxies for risk-seeking motives to investment in alternatives, both before and after controlling for risk constraints. Third, we explore whether shocks to expected spending can explain variation in the alternative-to-risky share.

4.1 Constraints on the Risky Share

In textbook portfolio choice models, a risk-seeking mechanism can affect the alternative-to-risky share only if the risky share is constrained. Empirically, however, it is difficult to determine whether a pension's risky share reflects a binding constraint or underlying risk preferences. To address this measurement issue, we study deviations $l_{pt} = \text{actual}_{pt}^{risky} - \text{target}_{pt}^{risky}$ of actual from target risky shares.[25] The idea is as follows. Market fluctuations will naturally move each pension's actual risky share from its target, after which pensions must rebalance to bring the two in line (Harvey et al., 2025). If a pension wants to increase its risky share but is constrained from doing so (e.g., by statute), it will want to rebalance quickly when its actual risky share falls below target. Conversely, it should be slower to rebalance when its actual risky share is above target because it prefers to take on the extra risk. Under this logic, persistently positive values of l_{pt} are indicative of a binding constraint.

To connect this measure of constraints to the growth of alternatives, we use the following cross-sectional regression:

$$\Delta\Omega_p = a + \theta \bar{l}_p + \varepsilon_p \tag{9}$$

where $\Delta\Omega_p$ is the change in the target alternative-to-risky share for pension p between 2002 and 2021 and $\bar{l}_p$ is the average l_{pt} over the same period. The coefficient of interest in the regression is θ. A large $\bar{l}_p$ is an indication that pension p has consistently faced binding risk constraints, in which case θ should be positive if this constraint induces them to shift more to alternatives.

Figure 6a visualizes the regression using a binned scatter plot of $\Delta\Omega_p$ and $\bar{l}_p$. The plot provides some indication that risk-constrained pensions have increased their alternative-to-risky share, though the correlation between the two is rather weak. The estimated coefficient in the plot equals 1.34 ($t = 1.78$) and the linear specification yields a fairly low R^2 of 4%. The regression coefficient indicates that a one-standard deviation increase in the average constraint is associated with a 3 pp increase in the alternative-to-risky share. The

[25] We plot and discuss the time-series properties of l_{pt} in Internet Appendix F.1.5.

standard deviation of the change in the alternative-to-risky share for this sample equals 16 pp.

One concern with this analysis is that deviations of actual from target risky shares reflect both market fluctuations and any potential risk constraints. To isolate the latter, we first regress l_{pt} on each year's portfolio return then compute $\bar{l}_p$ based on the regression residuals. The regression coefficient in this case is basically unchanged at 1.33 ($t = 1.73$).

Cash holdings provide another way to proxy for binding risk constraints, with the intuition being that a constrained pension will want to hold as little of its fixed income assets in cash as possible. Figure 6b tests this idea by replacing $\bar{l}_p$ in regression (9) with the contemporaneous change in each pension's target cash share. The figure offers some cross-sectional evidence that declines in cash holdings are associated with an increases in the alternative-to-risky share, but the correlation between the two is again rather weak. The regression yields an estimated coefficient equal to -0.74 ($t = -1.49$) and a low R^2 of 2%. In terms of magnitude, the regression coefficient implies that a one standard deviation decrease in the change in cash holdings is associated with a 2pp increase in the alternative-to-risky share.

As a complementary test, we instead regress the 2021 level of the alternative-to-risky share on the contemporaneous level of target cash share. Assuming cash holdings are a reasonable proxy for the tightness of risk constraints, then under the view that risk-constrained pensions are more likely to invest their risky assets in alternatives, we should find a negative correlation between the two. Consistent with this hypothesis, the coefficient from this regression equals -0.83 ($t = -2.18$). Still, the R^2 in the regression is 3% and the regression coefficient implies that a one-standard deviation decrease in the cash share is only associated with a 2 pp increase in the alternative-to-risky share. Thus, while we find some statistical evidence that risk-constrained pensions have shifted toward alternatives and away from public equities, the strength of this relationship is economically weak.

4.2 Risk-Seeking Motives

We next examine whether pensions with stronger incentives to take risk have shifted more aggressively out of public equities and into alternatives. Our baseline regression for this analysis is as follows:

$$\Delta\Omega_p = a + \beta\Delta X_p + \varepsilon_p, \tag{10}$$

where $\Delta\Omega_p$ is the change in the target alternative-to-risky share for pension p between 2002 and 2021. The explanatory variable ΔX_p is the contemporaneous change in one of several proxies for pension risk aversion. Standard errors are clustered by state for regressions run at the system level and are otherwise robust.

Our first proxy for risk-seeking motives is based on the idea that pension underfunding creates incentives for U.S. pensions to "reach-for-yield" via alternatives (Lu et al., 2019; Gillers, 2021), similar to those facing corporate equity holders when approaching default (Jensen and Meckling, 1976).[26] Figure A13a shows that these incentives have risen for U.S. pensions as interest rates have fallen over the last twenty years. At the beginning of the 2000s, U.S. pensions were more than fully funded according to standards set forth by Statement 25 of the Governmental Accounting Standards Board (GASB), but by 2021 their assets were only 75% of liabilities.[27] Figure A13b further shows that pension funding varies strongly in the cross-section, with some pension funding ratios sitting below 50% as of 2021.

Column (1) of Table 7 shows that changes in funding are only weakly negatively correlated with changes in alternative use. The estimate is not statistically significant ($p = 0.25$) and implies that moving from the 90th to 10th percentile of the change in funding is associated with a 4% increase in the alternative-to-risky share. Over this period, the actual spread between the 90th and 10th percentile of the change in the alternative-to-risky share was 40%. The R^2 of 1% in the regression is also low and we show in Internet Appendix F.1.1 that the poor model fit is not driven by a non-linear relationship between funding and alternative usage.

As Brown and Wilcox (2009), Novy-Marx and Rauh (2011), and others have pointed out, GASB 25 funding ratios overstate the true economic funding gap because liabilities are discounted using the expected rate of return on pension assets (asset hurdle rates), rather than more appropriate discount rates based on federal or local government debt. To handle this measurement issue, column (2) replaces GASB 25 funding ratios with those provided by the U.S. Bureau of Economic Analysis (BEA), who uses the yield-curve for AAA-rated corporate bonds to discount liabilities. The regression is run at the state level because pensions' BEA-funding ratios are only available at the state level. The point estimate in this case is effectively zero and the R^2 is once again low, suggesting that the measurement of funding is unlikely to drive its weak relationship with the adoption of alternatives.

[26] By reach-for-yield, we mean the propensity to take more risk as interest rates fall, though this is one of many ways that the term has been used in the literature. For example, Lian et al. (2018) show that investors may exhibit reach-for-yield behavior for behavioral reasons, as opposed to institutional or agency frictions.

[27] Under GASB 25 standards, funding ratios are computed from: (i) smoothed asset values that minimize the impact of short-term market fluctuations and (ii) liabilities that equal future benefits discounted using each plan's assumed long-run investment return.

Column (3) explores the related hypothesis that U.S. pensions have shifted to alternatives to meet sticky nominal return targets as interest rates have declined. To test this idea, we regress changes in the alternative-to-risky share on contemporaneous changes in each pension's hurdle rate. As with funding, the relationship is small in both statistical and economic terms and explains almost none of the variation in changes in the alternative-to-risky share.

A related hypothesis concerning the increased adoption of alternative investments centers on nominal return targets, or hurdle rates. Specifically, if U.S. pensions have fixed or relatively unchanging nominal hurdle rates, then a decrease in interest rates poses a challenge to meeting these targets. Consequently, pensions may seek to take on extra risk in search of higher expected returns. We test this proposition directly in column (3) of Table 7 by regressing changes in the alternative-to-risky share on contemporaneous changes in each pension's hurdle rate. If anything, the regression coefficient suggests a negative, not positive, correlation between alternative adoption and changes in liability discount rates. However, much like funding, the relationship is small in both statistical and economic terms and explains almost none of the variation in changes in the alternative-to-risky share.

Andonov et al. (2017) note that hurdle rates may be sticky precisely because lowering them causes U.S. pensions to appear more underfunded by GASB 25 accounting. They argue that this accounting rule creates incentives for pensions with a high fraction of retired members to take more risk. Underfunded pensions are typically required to amortize any unfunded accrued liabilities by making additional "catch-up" contributions, where the contribution amount depends on the funding deficit. Because more pensions with more retired members have larger accrued liabilities, their incentive to invest in risky assets is high because doing so would maintain their liability discount rate, lower their GASB 25 funding gap, and ultimately reduce the size of required contributions. Column (4) builds on Andonov et al. (2017) by testing whether pensions that have had more members retire are also those that have shifted more to alternatives. While there is a positive relationship between the two, the regression coefficient is not statistically different from zero.

One reason for these weak results may be that risk-seeking motives only affect the alternative-to-risky share if pensions cannot increase their risky share. For example, in mean-variance models, risk aversion only impacts the composition of the risky portfolio if pensions are leverage constrained. Columns (5)-(7) address this issue by controlling for the two measures of constraint slack developed in Section 4.1: (i) the average difference between actual and target risky shares; and (ii) the contemporaneous change in the target

cash share. The regressions are all run at the system level because this is the unit of observation for which our measures of constraint slack are constructed. In all cases, controlling for constraint slack does little to change the point estimates observed in columns (1), (3), and (4).

In summary, we find little evidence that funding status, return targets, or regulatory incentives account for why some U.S. pensions have increased their alternative-to-risky share more than others since the 2000s.[28]

Robustness In Internet Appendix F.1, we probe the robustness of this conclusion in a few ways. First, we document that the initial levels of the covariates in Table 7 also do not predict subsequent changes in the alternative-to-risky share. For example, it is not the case that pensions who were more underfunded in 2002 shifted more aggressively to alternatives between 2002 to 2021. There is also essentially no relationship between the change in the alternative-to-risky share and pension size or failure to make actuarial required contributions. Second, we show that the findings in Table 7 are not driven by our choice of a linear regression specification. Third, the relationship between the alternative-to-risky share and funding status remains weak when employing panel regressions in levels or focusing solely on levels in more recent data. Lastly, we document a similarly weak correlation between risk-seeking motives and changes in the alternative-to-risky share in the aftermath of the Global Financial Crisis of 2008.

4.3 The Covid-19 Pandemic

The preceding subsection linked several proxies for effective risk aversion (e.g., actuarial funding) to long-run changes in the alternative-to-risky share. We now develop a complementary research design that exploits the Covid-19 pandemic as a shock to current and expected spending. Since a smaller share of future obligations must be financed through portfolio returns, reductions in spending plausibly lower the incentive for plan sponsors to take investment risk. This corresponds to an increase in effective risk aversion in the mean-variance framework from Section 2.3. Under the view that the large alternative-to-risky shares observed at the end of our sample reflect leverage-constrained pensions, this increase in effective risk aversion should relax the shadow cost of the constraint and lead to a decline in the alternative-to-risky share.

The research design in this section also helps address limitations of using actuarial funding ratios to

[28]For completeness, we explore the role of funding and accounting-based mechanisms in explaining the increase of the overall risky share in Internet Appendix F.3 and find that they help explain why U.S. pensions have taken on more risk since the turn of the century, but their explanatory power is low.

measure effective risk aversion. The reason is that actuarial funding ratios are based on demographic assumptions (e.g., longevity) that are potentially endogenous and only react to spending shocks with a delay.[29] Our analysis proceeds in two steps: first, we argue that the shock of the pandemic plausibly induced cross-pension changes in effective risk aversion through its effect on spending needs. Second, we trace out pensions differentially adjusted their alternative-to-risky shares in response to the shock.

According to age-adjusted mortality rates from the Centers for Disease Control and Prevention (CDC), states differed widely in their exposure to the shock of the pandemic. For example, Hawaii averaged 25 deaths per 100,000 people in 2022, while Mississippi exceeded 100 deaths per 100,000. Our empirical design connects this variation to different state-level outcomes using the following regression:

$$\Delta Y_s = a + b \times \text{Covid Mortality Rate}_s + \varepsilon_s,$$

where ΔY_s is the change in a given outcome for state s relative to its pre-Covid level in 2019. In all specifications, we standardize the mortality rate to have a mean of zero and variance of one, and use heteroskedasticity-robust standard errors. Table 8 summarizes the results.

In column (1), the outcome variable is the change in age-65 life expectancy between 2019 and 2021, sourced from the CDC. Life expectancy changes are computed only through 2021 because 2022 data is not yet available. The purpose of this regression is to establish that the pandemic was indeed a longevity shock from the perspective of plan sponsors, one that plausibly reduced their future spending needs. The negative and statistically significant coefficient confirms this interpretation—states with higher COVID-19 mortality rates experienced larger declines in age-65 life expectancy. The magnitude of this effect is also meaningful: a one-standard deviation increase in mortality rates is associated with a 0.2-year decline in life expectancy, equivalent to roughly 60% of the cross-state standard deviation in life expectancy changes. From the perspective of a DB plan sponsor, a longevity shock of this magnitude is also likely to be economically meaningful (Kisser et al., 2012).

Column (2) further shows that states with high Covid mortality rates also experienced a reduction in the fraction of retired pension beneficiaries from 2019 to 2022. The statistically significant coefficient implies that a one-standard deviation increase in the Covid mortality rate lowered the fraction of retirees by about 0.3

[29] Actuaries have considerable discretion when setting mortality assumptions that underpin projects of future pension obligations. In addition, these assumptions are typically revised only every three to five years as part of scheduled experience studies, resulting in a sluggish adjustment of funding ratios to abrupt and salient demographic events such as the COVID-19 shock.

standard deviations. This result reinforces the notion that the pandemic lowered current and future spending needs for plan sponsors.

Columns (3) and (4) test whether the pandemic induced changes in the alternative-to-risky share.[30] In column (3), ΔY_s is the change in the target alternative-to-risky share for each state between 2019 and 2022. There is essentially zero correlation with Covid-19 mortality rates. Recognizing that policy portfolios may adjust slowly to expected spending shocks, column (4) extends the analysis to changes in the alternative-to-risky share from 2019 to 2023 and finds similar results.

In Internet Appendix F.2.1, we move beyond the Covid-19 shock and show that the alternative-to-risky share generally has little predictive power for future spending or cash-flow constraints—contrary to what one might expect if alternative allocations were shaped by forward-looking spending needs. Taken together, our findings suggest that spending needs are largely disconnected from alternative use. This may not be surprising, as there are likely offsetting forces at play. On the one hand, higher expected spending could push up the alternative-to-risky share through its impact on effective risk aversion, provided the risky share is also constrained. On the other hand, the same spending pressures might lead plans to prioritize liquidity, thereby discouraging allocations to illiquid alternatives. The weak empirical relationship we document may therefore reflect a balance of these competing considerations.

5 Aggregate Implications

Our results so far suggest that beliefs are a key driver of cross-sectional variation in the use of alternatives, while risk-seeking motives play a more limited role. In this section, we assess the relative importance of both channels for understanding the aggregate increase in the alternative-to-risky share. Our analysis proceeds in four steps. First, we simulate the mean-variance model from Section 2.3 to study the conditions under which reaching-for-yield behavior—equivalently, a decline in risk aversion—can match the evolution of the aggregate portfolio from 2001 to 2021. Second, we use the simulated model to assess how much beliefs about alpha would need to shift to explain the aggregate, incorporating the impact of consultants and peers that were estimated in Sections 3.1 and 3.3. Finally, we examine the behavior of U.S. endowments, U.S. corporate pensions, and U.K. corporate pensions to further shed light on the drivers of the alternative-to-risky share for U.S. public pensions.

[30] The rest of the paper uses the January 2023 release of the PPD, which is incomplete for fiscal years 2022 and 2023. For our analysis of the Covid-19 shock, we therefore use the December 2024 release of the PPD.

5.1 Simulating Reaching-for-Yield in the Model

Section 4 shows that risk-seeking motives and constraints explain only a small share of cross-pension variation in alternative use. This raises questions about their role in the aggregate trend, though this conclusion depends on how well we capture risk preferences and constraints. For example, funding may not perfectly capture a pension's effective risk aversion because the link between legally mandated contributions and underfunding may vary at the state or local level. Pension regulation is another potential confounding factor. In the late 1980s and 1990s, many state and local legislatures lifted restrictions on the ability of pensions to take risk, thereby decreasing the effective risk aversion of pensions in aggregate and potentially driving the rise in the alternative-to-risky share during the 2000s. Such complexities could introduce measurement error into our proxies for risk aversion or constraint slack, potentially attenuating their relationship with pension investment behavior in the cross-section.

To address these concerns, we use the mean-variance model from Section 2.3 to simulate how a decline in risk aversion coupled with a binding risk constraint would impact the composition of the aggregate pension portfolio. We do so for a wide range of beliefs that are held fixed throughout each simulation. This strikes us as a reasonable exercise given that, as previously mentioned, the large majority of U.S. pensions use mean-variance analysis as an input to portfolio construction.

We first assume that the aggregate pension portfolio was unconstrained in 2001 but that a minimum constraint on fixed income becomes binding by 2021. This means that the observed fixed income share in 2021 is the minimum required share ω_f^{min}. We then draw a random set of beliefs from the following uniform distributions: (i) expected excess returns on public equities, $\mu_E \sim U(0.02, 0.07)$; (ii) the variance of excess equity returns, $\sigma_E^2 \sim U(0.02, 0.09)$; (iii) risk-adjusted expected returns on alternatives relative to equities, $\alpha \sim U(0, 0.05)$; (iv) the CAPM-beta of alternatives relative to public equities, $\beta \sim U(0, 1.5)$. Realized excess returns on alternatives are therefore given by: $r_A - r_f = \alpha + \beta(r_E - r_f) + \eta_A$, where σ_η^2 is the idiosyncratic variance of alternatives. Together, these belief parameters fully define the expected excess return and variance-covariance matrix that determine optimal asset allocation. For each parameter draw, we set σ_η^2 to match the aggregate alternative-to-risky share Ω in 2001 according to Equation (2). Finally, we select γ_{2001} as the risk aversion that would also match the overall risky share in 2001, again assuming that pensions are not constrained. These last two steps ensure that each random belief set is consistent with the 2001 aggregate pension portfolio. We restrict the set of beliefs S^* so that $\sigma_\eta^2 > 0$, $\gamma_{2001} \geq 1$, and $\sigma_A^2 \leq 0.25$.

We then draw random belief sets until we reach $S^* = 100,000$ that satisfy these conditions.[31] Given this set of admissible initial beliefs, we then assume the portfolio becomes constrained in 2021 and solve for the new risk aversion γ_{2021} needed to generate the observed Ω in the data. Panels A and B in Table 9 summarize the simulation approach. In all cases, we impose a constraint that alternatives cannot be shorted, meaning $\omega_A \geq 0$.

Panel C of Table 9 contains a breakdown of our simulation outcomes. The headline result is striking: in 99.6% of simulations (99,621 out of 100,000), *no* change in risk aversion can account for the observed increase in the alternative-to-risky share. The intuition follows from revealed preference. In 2001, public equities accounted for the vast majority of the risky portfolio, which implies that pensions viewed them more favorably than alternatives at the time. Accordingly, in nearly every simulation, public equities have a higher expected return than alternatives in 2001. Consequently, for the vast majority of beliefs that match this initial composition (99,137 out of 100,000), pensions would want to shift even further towards public equities—*not* alternatives—when the constraint on the risky share binds.[32] In other words, $\Delta\Omega$ is counterfactually negative for these simulations.

There is a much smaller subset of simulations (484 out of 100,000) for which a decline in risk aversion would generate an increase in the alternative-to-risky share, but not enough to match the data before risk aversion hits its lower bound of 1. For this small subset of cases, the median alternative-to-risky share increases to a level of 28%, compared to 39% in the data. Panel C of Table 9 also shows that 0.4% of simulations (379 out of 100,000) succeed in matching the observed $\Delta\Omega$ with a decline in risk aversion. However, in Internet Appendix G.1, we show that the implied shadow cost of the risk constraint is implausibly high, as it implies that pensions would forgo over 700 basis points of returns per year to relax the constraint.

It is important to stress that the simulation exercise above does not rule out the possibility that U.S. public pensions are reaching for yield or facing binding risk constraints. On the contrary, the fact that the *risky* share has increased over the last twenty years is a clear indication that pension risk aversion has indeed declined. Rather, the model highlights that neither of these channels can sufficiently explain the increase in the *alternative-to-risky* share without a concurrent shift in beliefs. The cross-sectional evidence in Sections 4.1 and 4.2 further supports this interpretation, as does the positive evidence we provided in Section 3.

[31] See Figure A20 in the Internet Appendix for the histograms of admissible beliefs.

[32] More precisely, the mean-variance investor will shift toward public equities when their expected return exceeds that of alternatives, provided doing so does not add too much portfolio volatility on the margin.

5.2 The Role of Beliefs

Next, we modify the simulation framework from above to study how much beliefs would need to shift to explain the aggregate increase in the alternative-to-risky share. To do so, we no longer assume that pensions face constraints on the risky share and instead allow beliefs about alternatives to change. We start from the same set $S^* = 100,000$ of random initial beliefs and risk aversion parameters that match the aggregate pension portfolio in 2001. Within each simulation, we then solve for the increase in perceived alpha α_p of alternatives relative to public equities that is needed to match the alternative-to-risky share Ω in 2021. We then solve for the risk aversion γ_{2021} in 2021 needed to match the aggregate risky share.[33] Unsurprisingly, an increase in α_p can rationalize $\Delta\Omega$ for all $100,000$ initial beliefs. Figure A22a in the Internet Appendix shows how the implied increase in α_p varies across simulations. We winsorize the distribution at the 97.5th percentile to make the plot more readable. The average required increase in perceived alpha across simulations is 70 bps.[34]

The Direct Impact of Consultants In Internet Appendix G.2.2, we show how to further enrich this simulation by incorporating the causal impact of consultants estimated from Section 3.1. The basic idea is to express the total required change in alpha ($\Delta\alpha_p$) within each simulation as the sum of two components: $\Delta\alpha_p = \zeta\Delta\alpha_c + \Delta\xi_p$, where $\Delta\alpha_c$ is the observed change in consultant-perceived alpha in Figure 4a and $\Delta\xi_p$ is the change in the pension sector's private beliefs about the asset class. ζ is the causal influence of consultant beliefs on pension beliefs. In each simulation, it can be computed by feeding in the estimated elasticity b_1 of the alternative-to-risky share with respect to consultant beliefs from Section 3.1 into the mean-variance model from Section 2.3. ζ varies across simulations because it depends on b_1, as well as the initial beliefs and risk aversion parameters used in each simulation.

Figure 4a shows that the median consultant's perceived alpha rose by roughly 60 bps between 2001 and 2021. Given a value of ζ and setting $\Delta\alpha_c \approx 60$ bps, we can then back out the change in private pension beliefs $\Delta\xi_p$ needed to match the observed increase in the alternative-to-risky share. By construction, the implied $\Delta\xi_p$ in each simulation is therefore consistent with the composition of the aggregate pension portfolio in

[33] See Internet Appendix G.2 for the complete set of simulation results.

[34] Appendix G.2.1 conducts a complimentary exercise in which we increase the perceived alpha $\Delta\alpha$ by a fixed amount and compute the distribution of the implied change in the alternative-to-risky share $\Delta\Omega(\Delta\alpha)$ across simulations. This allows us to assess how the alternative-to-risky share responds to varying increases in perceived alpha, including those derived from estimates in the academic literature (e.g., Korteweg, 2019). The main takeaway is that, depending on initial beliefs in 2001, an increase in perceived alpha ranging from $\Delta\alpha = 20$-70 bps can reasonably account for the full rise in the aggregate alternative-to-risky share.

2001 and 2021, as well as the causal impact of consultants on the alternative-to-risky share estimated in Section 3.1. The average required increase in $\Delta\xi_p$ across simulations appears plausible and is about 60 basis points (see Figure A24). In other words, the causal estimates from Section 3.1 suggest that about 12% of the aggregate trend in the alternative-to-risky share can be attributed to consultants becoming more bullish about alternatives.

Peer Effects Our results from Section 3.3 further imply that the required increase of $\Delta\xi_p \approx 60$ bps in pension-specific beliefs could be partly driven by peer effects. For example, if one pension has an idiosyncratic belief shock about alpha, the existence of peer effects means that the shock will impact the behavior of other pensions and thus the aggregate portfolio composition. In Internet Appendix G.2.3, we use standard techniques from network economics to translate our estimates of peer effects into multipliers that measure how the peer network amplifies belief shocks in aggregate (e.g., Leontief, 1986; Acemoglu et al., 2016; Herskovic et al., 2020). Our analysis suggests that amplification by the peer network can account for about 20% of the increase in pension-specific beliefs needed to match the aggregate alternative-to-risky share.

5.3 Evidence from Other Institutions

We conclude our aggregate analysis by exploring the idea that the perceived alpha of alternatives has increased for all types of investors, not just public pensions. We do so by documenting aggregate trends in the portfolio composition of DB pensions sponsored by U.S. corporations, U.K. corporations, and U.S. endowments. Figure 7a plots the alternative-to-risky share for each institution through time, revealing a clear and common upward trend for all institutions. In contrast, Figure 7b shows that there is no such common trend in the overall risky share. Despite starting at relatively similar levels in the mid-2000s, the risky share for corporate pensions in the U.K. and U.S. has sharply declined, whereas it has slightly increased for U.S. endowments and public pensions. U.K. pensions provide the most striking example of these diverging trends. From 2004 to 2021, their alternative-to-risky share more than quadrupled from 11% to 50% while their risky share more than halved from 69% to 31%.

These findings are closely related to the work of Ivashina and Lerner (2018), who document that public- and private-sector DB pensions have both increased their overall share of alternatives since 2008. We extend their results by showing that this trend cannot be explained by a common desire of all institutions to take on more risk and has instead largely occurred through a change in the *composition* of risky investments.

Beliefs about asset returns are a natural explanation for this fact, especially from the perspective of canonical portfolio choice models. For instance, it seems natural to expect that investors have updated their beliefs about the risk-reward properties of alternatives as the asset class has matured. It is also conceivable that the structural forces behind the decline in interest rates have simultaneously impacted the perceived benefits of alternatives relative to public equities. These types of belief-based explanations are appealing because they can account for why institutions that vary widely in governance structure, regulation, funding, and economic function (e.g,. endowments vs. DB pensions) have all shifted the composition of their risky investments – but not their overall risky share – in the same way.

6 Other Explanations

In this section, we consider two explanations for the data that fall outside the mean-variance framework developed in Section 2.3. The first involves the supply side and the second is an agency-based interpretation of our findings.

6.1 Supply-Side Considerations

Up to this point, we have focused on demand-side explanations for the rise of alternatives, but supply-side developments may also be important. For example, the growth of the private capital industry has broadened the investment global market portfolio by giving institutions greater access to privately held firms, real estate, and infrastructure. While this sort of supply-side expansion may help explain the aggregate trend, it is less likely to account for the wide cross-sectional variation in the alternative-to-risky share that persists today.

One related hypothesis is that observed shift toward alternatives by U.S. pensions could simply reflect their effort to hold an evolving investable global market portfolio (GMP). We test this idea by comparing the composition of the aggregate pension portfolio to State Street's GMP, which provides benchmark weights for a number of asset classes based on their market capitalization.[35] We focus on the composition of the risky portfolio (i.e., excluding fixed income) because pensions may have different risk aversion than the market, though our conclusions are robust to using raw portfolio shares.

Figure 8a plots the time-series of the alternative-to-risky share in the GMP, along with the target alternative-to-risky share of the U.S. public pension sector. The plot confirms that the alternative-to-risky share of the

[35] Internet Appendix H.2 contains additional details on State Street's GMP, including the adjustments we make to better align it with the asset class definitions in the PPD.

GMP has indeed increased over the last twenty years, rising from just over 10 pp at the turn of the century to approximately 23 pp in 2021. Though the alternative-to-risky share for pensions appears to track the GMP quite well until 2010, the two diverge sharply thereafter. As of 2021, pensions were overweight alternatives in their risky portfolio by roughly 17 pp relative to the GMP.

The alternative-to-risky share is a somewhat crude way to assess diversification because it ignores heterogeneity in different types of risky investments. To illustrate this point more clearly, suppose that the growth of private equity and the decline of public equity causes the alternative-to-risky to increase by 10 pp. If pensions respond by selling out of public equities and increasing their (risky) allocation to real estate by 10 pp, their alternative-to-risky share will also rise by 10 pp. However, this shift would clearly not reflect a desire by pensions to hold the market, as the underlying composition of their risky portfolio would differ significantly from the GMP.

To account for heterogeneity in different types of risky investment, we therefore construct the following diversification metric based on the work of Cremers and Petajisto (2009):

$$D_t = \frac{1}{2} \sum_{j=1}^{J} |\Omega_{j,t} - G_{j,t}|, \tag{11}$$

where $\Omega_{j,t}$ is the share of asset class j in the risky portfolio of U.S. public pensions at t and $G_{j,t}$ is the contemporaneous share in the GMP. The J asset classes we consider are based on the PPD and include public equities, private equity, real estate, hedge funds, and commodities.[36] Cremers and Petajisto (2009) originally introduced D_t under the label "Active Share" to study the difference between mutual fund holdings and their benchmarks. To avoid confusion with the active portfolio changes analyzed in Calvet et al. (2009)—which we use in our study of experience effects in Section 3.2—we instead refer to D_t as "distance". By construction, when $D_t = 0$, the risky portfolio of the U.S. pension sector lines up exactly with that of the GMP, at least at the asset class level. In contrast, large values of D_t indicate that pensions are far from holding the market.

Figure 8b presents the time-series of D_t, where the blue line represents the version based on target pension portfolio weights, and the red line represents the version based on actual pension portfolio weights. The plot reveals a striking trend: over the last twenty years, the risky portfolio of U.S. public pensions has steadily become *less* diversified relative to the GMP, even during the early part of the 2000s. This

[36] Consistent with our treatment of the PPD, private equity includes private equity and private credit.

observation contradicts the idea that the aggregate rise of alternatives was primarily driven by a desire to hold the market portfolio.[37]

The cross-section reveals a similar story. By the end of our sample in 2021, the vast majority of pensions were overweight alternatives in their risky portfolios, with more than 60% having an alternative-to-risky share that exceeds the GMP by at least 10 pp. Additionally, in Internet Appendix H.2, we show that the decline in diversification—reflected in rising values of D_t—is largely driven by an increase in the alternative-to-risky shares across pensions. There is also little evidence that the least diversified pensions in the early 2000s were the ones that shifted most aggressively toward alternatives. Taken together, these findings suggest that the rise of alternatives among U.S. pensions is unlikely to have been driven primarily by a desire to match the composition of the global market portfolio.

6.2 An Agency Interpretation

While our findings support the idea that pensions and their consultants increasingly believe that alternatives deliver alpha relative to public equities, it is also possible that stated beliefs reflect deeper agency frictions. For instance, consultants may publish bullish CMAs on alternatives to steer clients toward asset classes that require additional high-fee services, such as manager selection and due diligence (Andonov et al., 2023). Similarly, both pensions and consultants may prefer alternative assets because they are infrequently marked-to-market, which complicates risk-adjustment and can give the appearance of outperformance (Lo, 2001; Jurek and Stafford, 2015; Couts et al., 2020; Stafford, 2022). Though formal academic work on this mechanism is limited, the idea that institutional investors are attracted to the illiquidity and opacity of alternative assets has been widely discussed by practitioners and market observers (Asness, 2019; Ilmanen et al., 2020).

Our results place useful restrictions on the underlying nature of agency problems that are potentially reflected in stated beliefs. To explain the aggregate rise of alternatives, the incentive or capacity to conceal risk would need to have increased not only among U.S. pensions, but also across a wide range of global public- and private-sector institutions (Section 5.3). One possible microfoundation is that institutional investors broadly value financial products that smooth returns—for reasons that transcend local regulatory and accounting regimes—and the maturation of alternative asset classes has made it easier to act on that preference. From this perspective, the rise of alternatives would reflect a response by the financial services

[37] Figure A26a in the Internet Appendix provides an additional breakdown of the components driving the growth in D_t.

industry to meet demand for return-smoothing products. This view would also help explain why many institutional investors are now overweight alternatives relative to the global market (Section 6.1).

An expansion in the availability of investment products that look good on paper cannot, however, account for the wide cross-sectional dispersion in alternative use and adoption across pensions (Section 2). To fully rationalize this variation, the desire for smoothed returns must also be largely orthogonal to a long list of observable pension characteristics, including pension size, funding status, liability discount rates (return targets), liquidity, spending needs, beneficiary demographics, board structure (Andonov et al., 2018), home bias (Hochberg and Rauh, 2012), and state-level trends (Sections 3.1.3 and 4.2).[38] Given that pensions vary considerably in their adoption of alternatives, this incentive would also need to have increased over time for some pensions but not others.

Although we cannot easily identify or measure a specific friction that satisfies all of the criteria discussed above, we recognize the inherent difficulty of using observational data to determine whether beliefs reflect genuine convictions or are shaped by deeper agency problems. Nevertheless, several pieces of evidence help to at least to rule in a meaningful role for genuine beliefs. For one, as we document in Section 2.2.4, pensions differ substantially in how they allocate within alternative asset classes, even those that are similarly illiquid. This fact is difficult to reconcile with a generic return-smoothing motive and is more naturally explained by beliefs.[39]

More directly, cross-sectional variation in the use of specific alternatives lines up closely with consultant beliefs. For example, Internet Appendix H.1.1 shows that pensions are more likely to invest in real assets over private equity—both of which presumably conceal risk equally well–when their consultant reports real assets as having a higher alpha. This result holds true even after accounting for any time-varying pension characteristics that may drive a desire to take on risk or invest in unmarked assets. Taken together, these findings point to an important role for genuine beliefs—one that is distinct from return-smoothing motives—in explaining the rise of alternatives. Additional discussion of return-smoothing motives appears in Internet Appendix H.1.1.

[38] We show in Internet Appendices H.1.2 and H.1.3 that board composition and home bias are unrelated to alternative use.

[39] Internet Appendix E.2 further shows heterogeneity in the use of specific alternatives is largely unexplained by proxies for risk-seeking motives and lines up much more strongly with consultant identity.

7 Conclusion

Since the early 2000s, public pensions in the United States have substantially altered the composition of their risky investments, actively reallocating away from public equities and into alternative investments like private equity, real estate, and hedge funds. This paper argues in favor of a relatively simple explanation of this trend: pensions increasingly believe alternatives earn alpha relative to public equities. These beliefs are shaped by consultants, peers, and experience during the 1990s. Other factors, including risk-seeking motives, spending needs, and governance appear to play a more limited role. There is also little evidence that pensions are attempting to match the global investable market portfolio, as they are now heavily overweight alternatives. Agency-based interpretations centered on the illiquidity of alternatives are possible, especially in aggregate, though heterogeneity in the use of specific subcategories of alternatives still points to a meaningful role for genuine beliefs about alpha.

A natural question that emerges from our analysis is whether pension and consultant beliefs about alpha are rational.[40] This is difficult to answer definitively given the inherent challenge of measuring the alpha of alternative investments, especially illiquid ones (e.g., Gupta and Van Nieuwerburgh, 2021; Korteweg, 2023; Brown et al., 2025).[41] In Internet Appendix E.3, we take a preliminary step toward addressing this question, though the evidence is mixed at best and does not permit strong conclusions. On the one hand, consultant beliefs from CMA data are not wildly out of line with estimates from historical data or the academic literature. However, the associated confidence intervals are wide—perhaps helping to explain why pensions appear to disagree so much. On the other hand, pensions with high alternative-to-risky shares do not clearly outperform those with lower shares, either within their alternatives portfolios or at the overall system level, as one might expect under fully rational expectations. Thus, our main takeaway from this preliminary analysis is that more work is needed to assess whether beliefs about alpha are rational. This question is critical for assessing the welfare implications of alternatives for pension beneficiaries, especially given the costs and complexity of investing in this asset class (Metrick and Yasuda (2010); Phalippou, Rauch, and Umber (2018); Begenau and Siriwardane (2022)).

[40] We are agnostic about belief rationality—what matters for our purposes is that beliefs help explain variation in portfolio choice.

[41] In recent work, Korteweg et al. (2024) evaluate PE investments made by pensions, and argue that the alpha earned by the representative pension funds is approximately zero. In the cross-section, they document that the PE returns of some plans with greater risk-seeking incentives reflect compensation for risk, not alpha.

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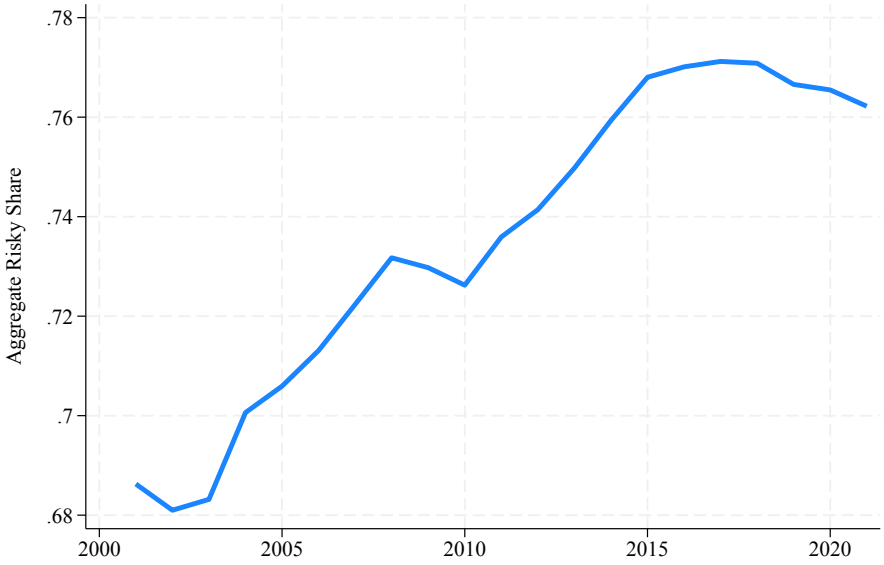
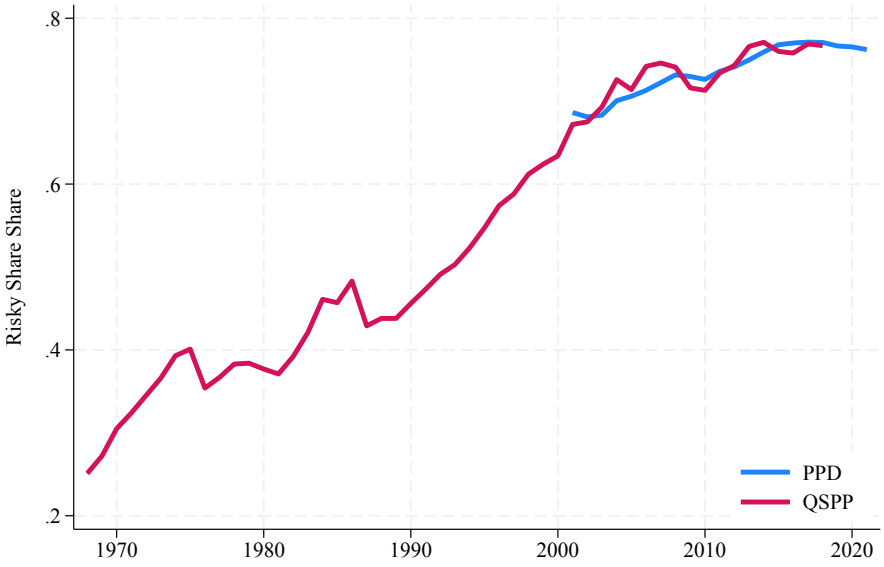
Figure 1: The Risky Share

(a) Aggregate, Post-2000

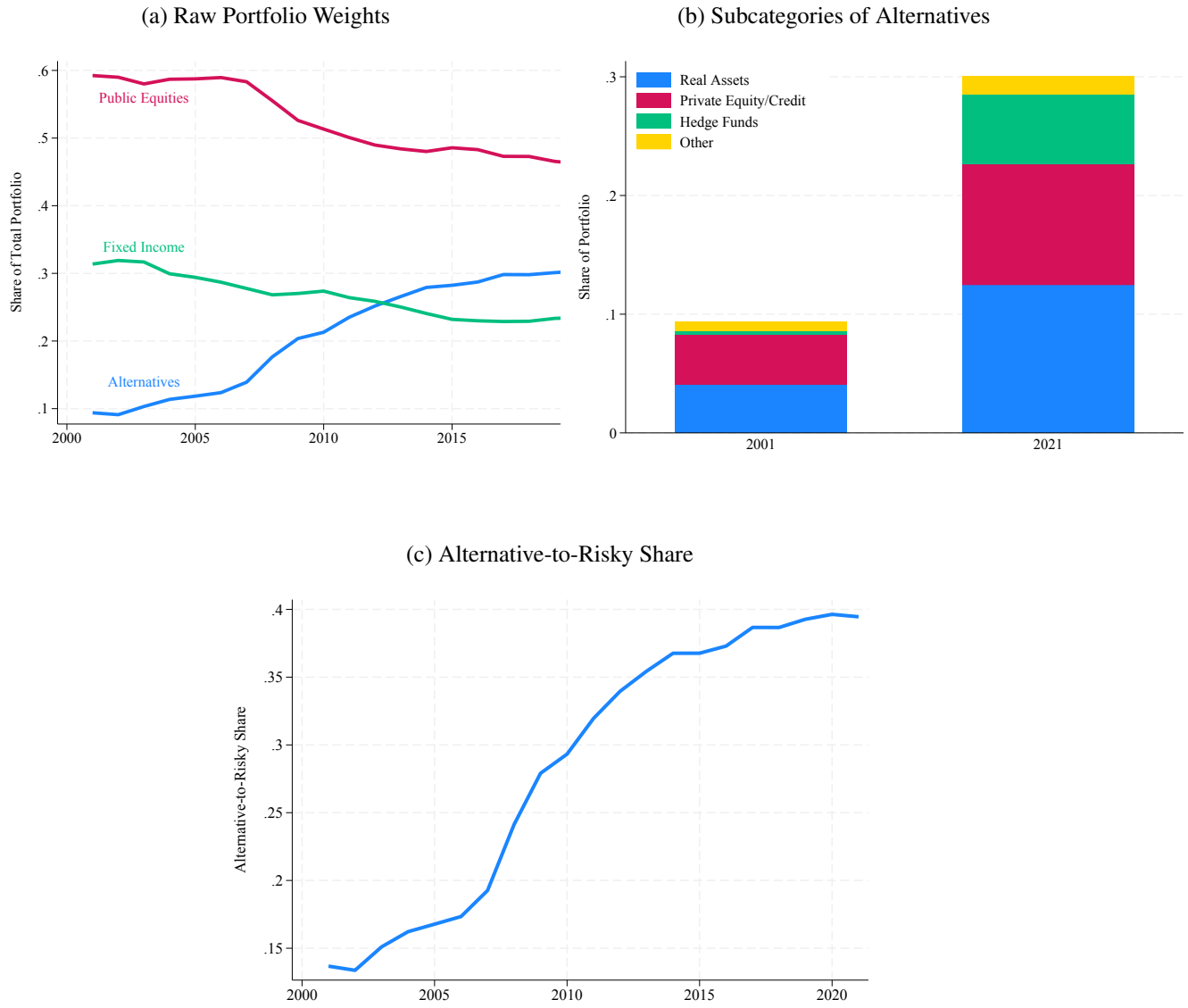


(b) Aggregate, Post-1968



Notes: Panel (a) plots the aggregate target risky share for U.S. public pensions based on data from the PPD, where the risky share is defined as any holding outside of fixed income and cash. Panel (b) adds a longer-history of the risky share using data from the U.S. Census Bureau's Quarterly Survey of Public Pensions (QSPP). The risky share in the QSPP similarly excludes fixed income and cash but is instead based on actual, not target weights. See Section 2 for complete details.

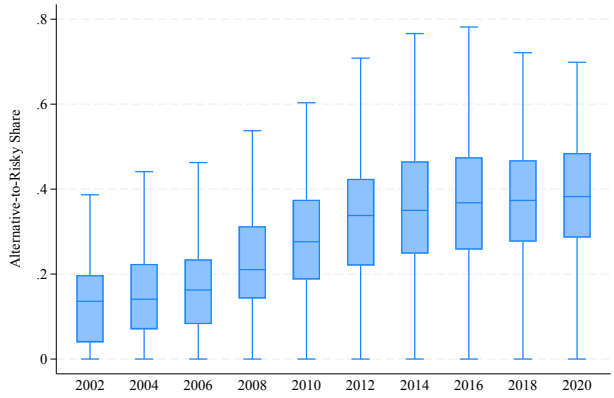
Figure 2: The Composition of the Aggregate Risky Portfolio



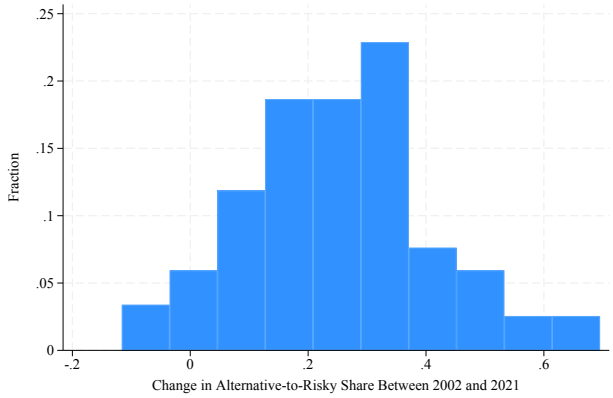
Notes: Panel (a) of the figure plots the target share of public equities, alternatives, and fixed income (including cash) for the aggregate U.S. public pension portfolio. Panel (b) shows aggregate target portfolio shares of different categories of alternatives for 2001 and 2021. Panel (c) plots the target share of alternatives in the risky portfolio. All data are based on the PPD. Risky investments are defined as any holding outside of fixed income and cash.

Figure 3: Alternative Use in the Cross-Section of Pensions

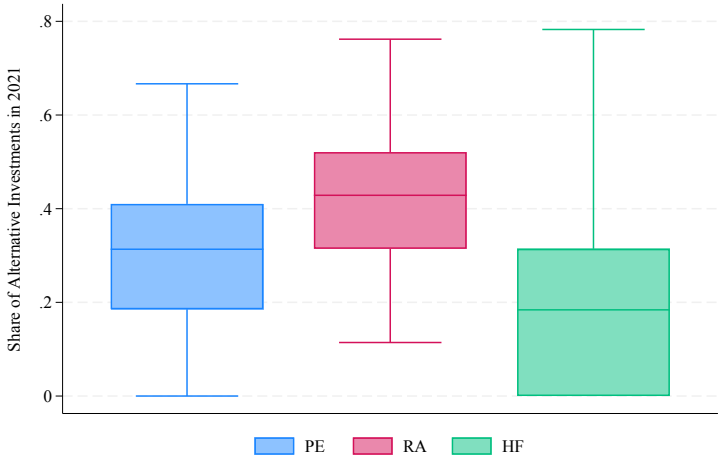
(a) The Cross-Section of the Alternative-to-Risky Share



(b) Change from 2002 to 2021



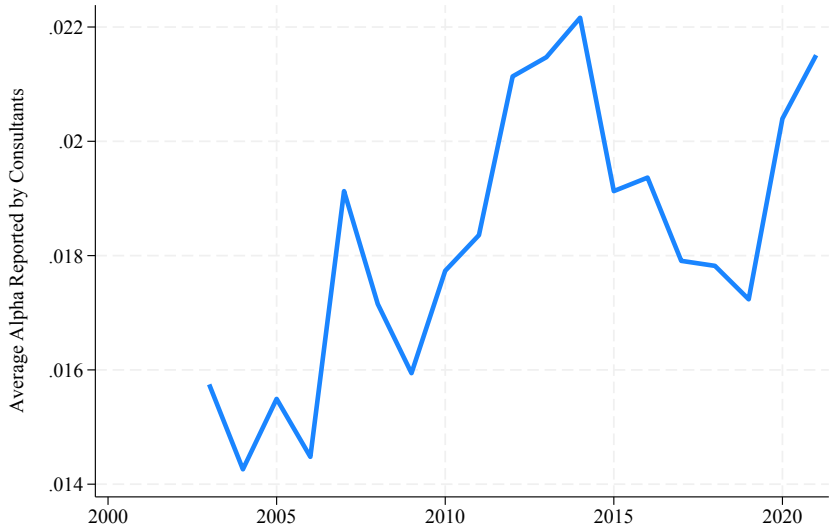
(c) Cross-Pension Heterogeneity in Specific Types of Alternatives



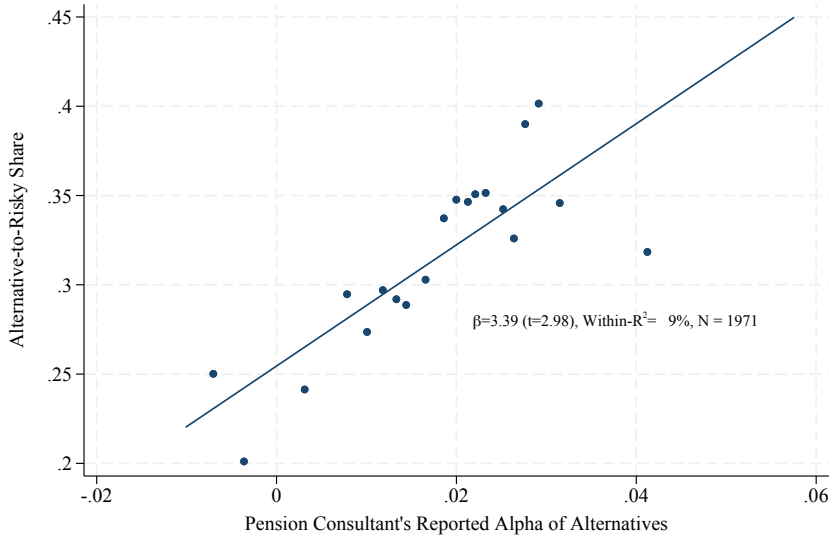
Notes: Panel (a) of the figure depicts the distribution of the target alternative-to-risky share across pension systems through time. Each box plot summarizes the distribution for the corresponding year on the *x*-axis. Only even years are plotted to make the graph more readable. Panel (b) plots the distribution of the change in the target alternative-to-risky share across U.S. pension systems between 2002 and 2021. Panel (c) visualizes heterogeneity in the composition of alternatives across pensions. For each category, the plot shows the distribution of its share within the total alternatives portfolio in 2021.All data are from the PPD.

Figure 4: Consultant-Reported Beliefs about the Alpha of Alternatives

(a) Median Alpha Over Time

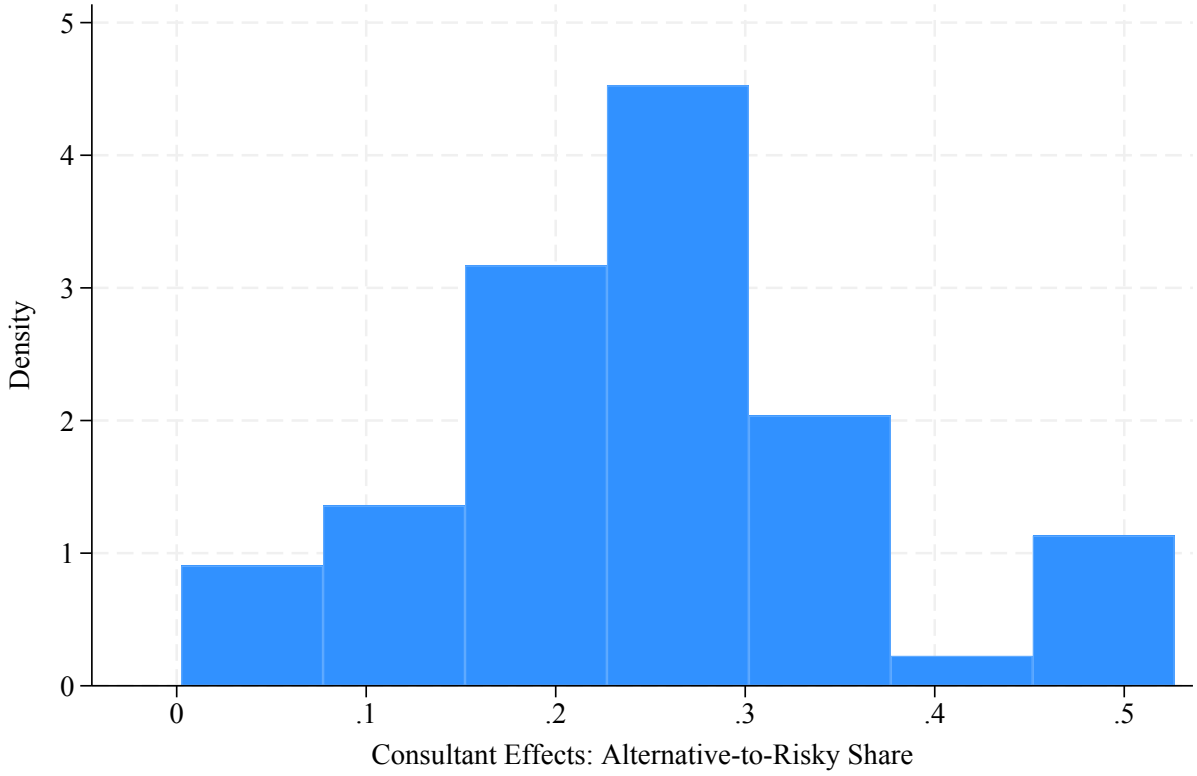


(b) Clients' Alternative-to-Risky Share



Notes: Each year in Panel (a) shows the median perceived alpha of alternatives relative to large-cap US equities across consultants. The data for the plot are based on hand-collected capital market assumption (CMA) reports that were obtained under confidentiality agreements. Panel (b) is a binscatter of each pension's target alternative-to-risky share against its consultant's reported alpha, after controlling for state-by-year fixed effects, funding status, (log) size, and hurdle rates. The reported t-statistic in panel (b) based on standard errors that are clustered by consultant and pension. See Section 3.1.2 for details.

Figure 5: Consultant Fixed Effects



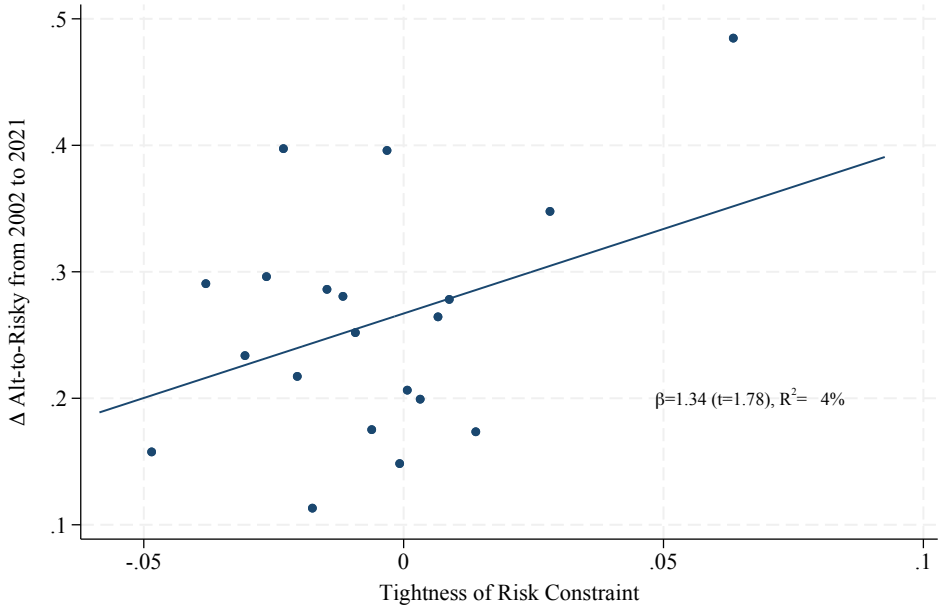
Notes: This figure shows the distribution of consultant effects λ_c estimated from the following regression:

$$\Omega_{p(c),t} = \lambda_t + \sum_k \beta_t^k \times X_{pt}^k + \lambda_c + \varepsilon_{pct}$$

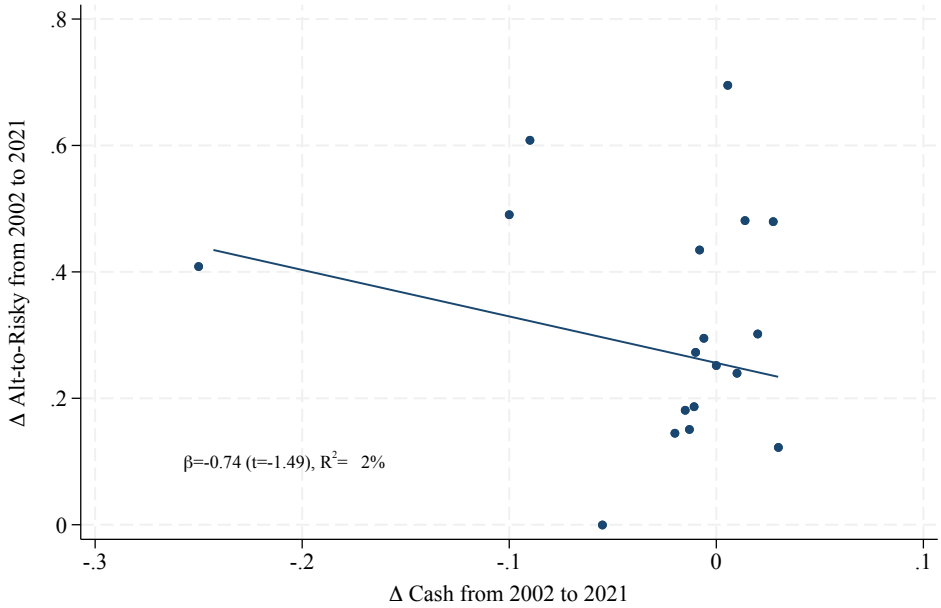
where $\Omega_{p(c),t}$ is the target alternative-to-risky share for pension p matched with consultant c in fiscal year t. λ_t is a time fixed effect and X_{pt}^k is the set of covariates including (log) size, GASB 25 funding, asset hurdle rates, the ratio of required actuarial contributions to payroll, and the ratio of administrative expenses to payroll. The estimated consultant fixed effects λ_c are then shrunk towards their mean using an empirical Bayes estimate (Casella, 1992) to account for sampling error. The average target alternative-to-risky share is added back to the fixed effects to facilitate interpretation.

Figure 6: Alternative Adoption and Risk Constraints

(a) Actual-Target



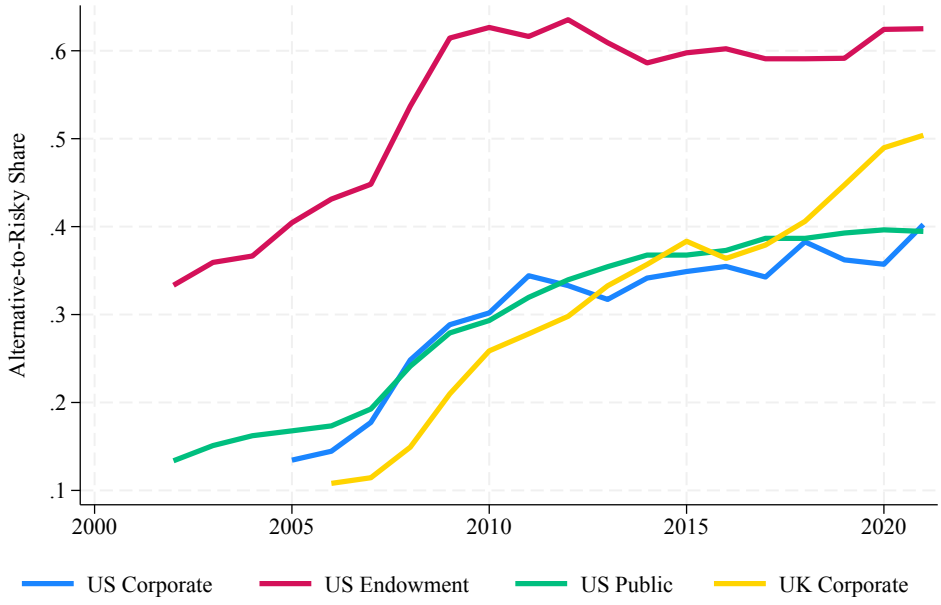
(b) Cash Share



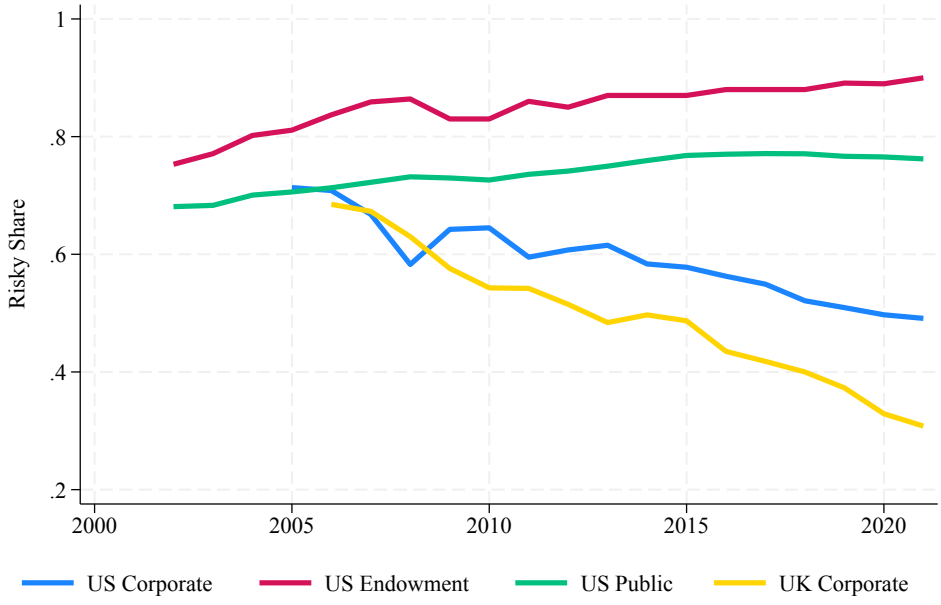
Notes: Panels (a) and (b) of this figure respectively show a binscatter of the change in the target alternative-to-risky share from 2002 to 2021 against the average tightness of portfolio constraints over the same period and the change in the target cash share. In panel (a), constraint slack in a given year t for a pension p is defined as the difference l_{pt} between its actual and its target risky share. Higher values of $\bar{l}_p$ indicate a tighter constraint. Both plots reflects only the sample of pensions for which both actual and target shares are available. The OLS regression line is also shown in both plots, with standard errors clustered by state. Data are at the pension-system level. See Section 4.1 for details.

Figure 7: Portfolio Behavior Across Institutions

(a) Alternative-to-Risky Share



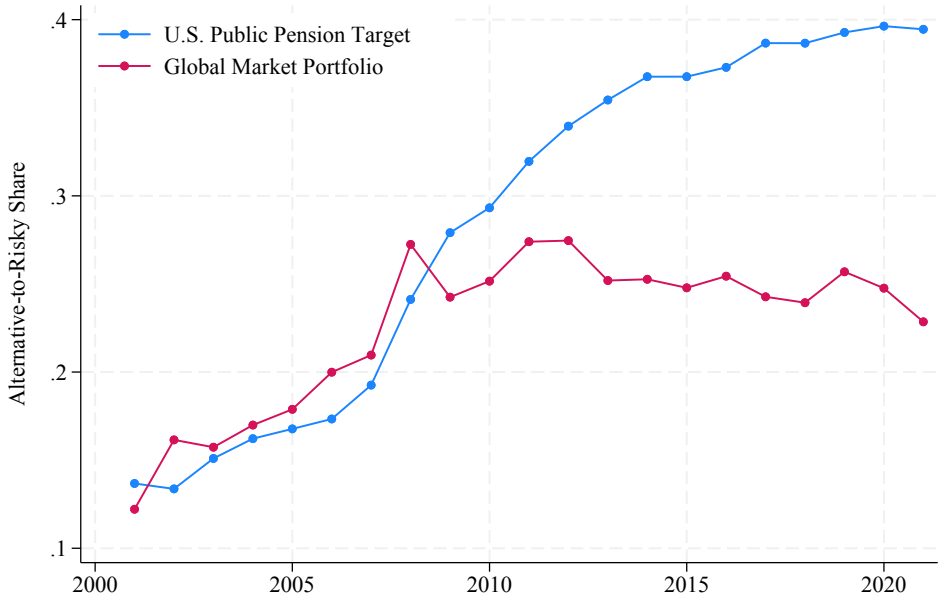
(b) Risky Share



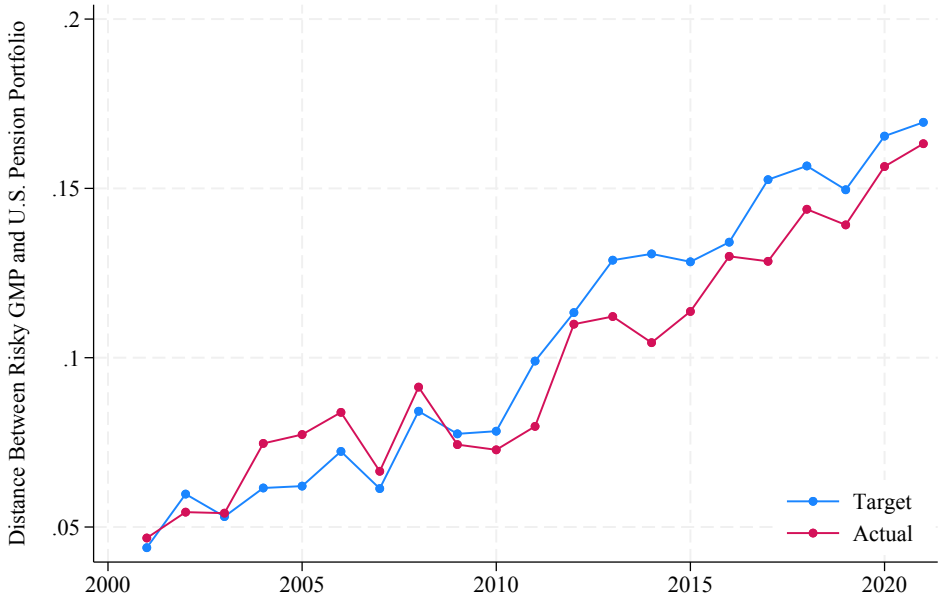
Notes: Panels (a) and (b) of the panel plot the alternative-to-risky and risky shares, respectively, of different institutions through time. U.S. Endowment data is from NACUBO historical endowment studies, U.K. corporate defined-benefit pension data is from the U.K. Pension Protection Authority, and U.S. corporate pension data is from the Milliman Corporate 100 report. Actual portfolio shares are used for U.S. endowments, U.S. corporate pensions, and U.K. corporate pensions, whereas target portfolio shares are used for U.S. public pensions. See Section 5.3 and Section A of the Internet Appendix for more details on the data.

Figure 8: Diversification Motives

(a) Global Alternative-to-Risky Share



(b) Distance from Global Market Portfolio



Notes: Panel (a) plots the alternative-to-risky share in the global market portfolio (GMP) through time, along with the aggregate target alternative-to-risky share for the U.S. public pension sector. Panel (b) plots the distance D_t between the risky portfolio of the U.S. public pension sector and that of the GMP, with distance defined using the "Active Share" metric from Cremers and Petajisto (2009). The blue line in panel (b) measures D_t using target pension portfolio shares, whereas the red line uses actual pension portfolio shares. Data for the GMP is taken from State Street, with adjustments made for REITs and hedge funds. See Section 6.1 and Internet Appendix H.2 for complete details of the data. The risky portfolio includes all investments outside of fixed income.

Table 1: National Summary Statistics

	Subsample			
	2001-2005	2006-2010	2011-2015	2016-2021
Number of Systems	165	190	202	208
Members (mm)	21	24	25	27
Percent Retired	28	31	35	37
AUM ($ bn)	2,120	2,633	3,154	4,065
GASB 25 Funding (%)	91	81	73	73
Assumed Asset Return (%)	8.0	7.9	7.6	7.2
Annual Investment Return (%)	5.3	6.2	9.1	10.0
National Coverage (%)				
Public DB Pensions	87	90	92	91
All Private and Public Pensions	25	25	24	22
Portfolio Composition (%)				
Fixed Income	30	27	25	23
Public Equities	59	54	49	47
Alternatives	11	18	27	30

Notes: This table provides national-level summary statistics for defined benefit public pensions in the United States from 2001 to 2021. Individual pension plans are aggregated to pension systems when the assets of multiple plans are legally pooled and managed together. GASB 25 funding is defined as the ratio of actuarial assets to liabilities, where liabilities are computed by discounting future promised benefits using each plan's assumed long-run rate of return on assets. The rows listed below *National Coverage* report the average annual ratio of assets in PPD to defined benefit pension assets listed in the U.S. Census Bureau's Annual Survey of Public Pensions and total public and private-sector pension assets listed in the Flow of Funds. The rows listed below *Portfolio Composition* show the target percent of the aggregate public pension portfolio invested in fixed income (including cash), public equities, and alternatives, respectively. Alternatives encompass investments in hedge funds, private equity, private debt, and real assets (e.g., real estate private equity). Data are based primarily on the Public Plans Database (PPD) that is maintained by the Center for Retirement Research at Boston College.

Table 2: Changes in the Average Consultant-Perceived Alpha

	$\bar{\alpha}$			$\bar{\mu}_A$			$\bar{\beta}$		
	2003	2021	Δ	2003	2021	Δ	2003	2021	Δ
All Alternatives	157	215	58	429	485	57	0.56	0.61	0.05
PE	92	180	88	660	705	45	1.18	1.19	0.02
RE	236	275	39	336	429	93	0.24	0.34	0.10
HF	144	190	46	290	322	32	0.26	0.29	0.03
Public Equities	-	-	-	467	437	-29	-	-	-

Notes: This table summarizes changes between 2003 and 2021 in the average consultant-perceived α, expected excess return, and β of all alternatives, as well as private equity (PE), real estate (RE), and hedge funds (HF). The average α of all alternatives is computed by first averaging the perceived α for each asset class across consultants, then averaging across the three asset classes. The other quantities are computed in a similar manner. Only consultants with at least ten years of data are included in the analysis.

Table 3: Consultant Beliefs and Pension Allocations

	Alternative-to-Risky Share$_{p,t}$			
	(1)	(2)	(3)	(4)
$\alpha_{c(p),t}$	3.39**	3.45**	3.27**	4.01**
	(2.98)	(3.19)	(4.24)	(5.27)
Estimation Type	OLS	OLS	IV	IV
Controls	No	Yes	Yes	Yes
Fixed-Effects	Time	Time	Time	Time
$p_{wild}(\beta = 0)$	0.01	0.01	0.01	0.01
IV Location Date			t	2005
First-Stage F			84.62	20.04
β_{IV}/β_{OLS}			0.95	1.20
Within-R^2	0.09	0.13		
N	1,971	1,957	1,957	1,637

Notes: This table summarizes linear regressions of pension p's target alternative-to-risky share in year t on the reported alpha of alternatives of pension p's consultant in year t. Pension controls include the level of pension funding, (log) size, hurdle rate, the ratio of required actuarial contributions to payroll, the ratio of administrative expenses to payroll, the cash share, annual return of the fund, and the residualized difference between each pension's target and actual risky share. Standard errors are clustered at the consultant and year levels, and t-statistics are listed below point estimates in parentheses. ** indicates a p-value of 0.05 and * indicates a p-value of 0.10.

Table 4: Consultants and the Composition of Risky Investments

			Fixed Effects							
	Dep. Variable:	Controls	Time	State-Time	Consultant	F	p	Adj. R^2	C	N
(1)	Alternatives	x	x					0.33		3,088
(2)	Alternatives	x	x		x	18.52	0.00	0.50	57	3,083
(3)	Alternatives	x		x				0.51		2,691
(4)	Alternatives	x		x	x	12.95	0.00	0.63	54	2,685
(5)	Private Equity/Credit	x	x					0.19		3,088
(6)	Private Equity/Credit	x	x		x	17.04	0.00	0.38	57	3,083
(7)	Private Equity/Credit	x		x				0.33		2,691
(8)	Private Equity/Credit	x		x	x	13.06	0.00	0.49	54	2,685
(9)	Hedge Funds	x	x					0.15		3,088
(10)	Hedge Funds	x	x		x	11.41	0.00	0.29	57	3,083
(11)	Hedge Funds	x		x				0.30		2,691
(12)	Hedge Funds	x		x	x	9.21	0.00	0.43	54	2,685
(13)	Real Assets	x	x					0.17		3,088
(14)	Real Assets	x	x		x	16.72	0.00	0.36	57	3,083
(15)	Real Assets	x		x				0.31		2,691
(16)	Real Assets	x		x	x	13.52	0.00	0.48	54	2,685

Notes: This table shows fixed effects regressions of the following form:

$$\Omega_{p(c),t} = \lambda_t + \lambda_c + \Gamma X_{p,t} + \varepsilon_{p(c),t}$$

where $\Omega_{p(c),t}$ is one of several target asset allocations for pension system p matched with consultant c in fiscal year t. λ_t is a year or year-by-state fixed effect, $X_{p,t}$ is a vector of characteristics of pension p in year t, and λ_c is a consultant fixed effect. Control variables include (log) size, GASB 25 funding, asset hurdle rates, the ratio of required actuarial contributions to payroll, and the ratio of administrative expenses to payroll. Some regressions also include a pension fixed effect. The listed F-statistic is the result of testing the null hypothesis that the consultant effects λ_c are equal to each other. C is the number of consultant effects that are included in the test. Sample sizes differ across models because we drop singleton groups for any included fixed effects. All allocations are relative to the overall target share of risky investments.

Table 5: Experience in the 1990s

	Δ Alt-to-Risky$_{2002\to2021}$				Δ Risky$_{2002\to2021}$
	(1)	(2)	(3)	(4)	(5)
Δ Equity$_{1996\to2002}$	0.50**	0.75**	0.87**	0.93**	0.24
	(3.06)	(2.98)	(3.43)	(3.14)	(1.36)
Δ Equity$_{1990\to2002}$		-0.09	-0.24	-0.27	-0.06
		(-0.37)	(-0.98)	(-0.86)	(-0.32)
Fraction of Restricted Assets$_{1998}$				0.01	0.01
				(0.12)	(0.20)
Financial Sophistication$_{1998}$				0.04	0.07*
				(0.53)	(1.90)
Political Board$_{2002}$				-0.07	0.02
				(-1.03)	(0.24)
Controls	No	No	No	Yes	Yes
1990 Method		Actual	Active Change	Active Change	Active Change
R^2	0.16	0.19	0.20	0.26	0.37
N	47	46	46	46	46

Notes: This table shows OLS regressions of the following form:

$$\Delta\Omega_{s,2002\to2021} = c + \beta_1 \Delta\text{Equity}_{s,1996\to2002} + \beta_2 \Delta\text{Equity}_{s,1990\to2002} + \Gamma X_s + \varepsilon_s.$$

The dependent variable in columns (1)-(4) is the state-level change in the target alternative-to-risky share and in column (5) it is the change in the target risky share. Both are computed using data from the PPD between 2002 and 2021. The regressor $\Delta\text{Equity}_{s,1996\to2002}$ is the change in the target share of public equities between 1996 and 2002 computed using target shares from the PENDAT 1996 survey and target shares from PPD in 2002. In the 1996 PENDAT and PPD datasets, missing target shares are replaced by actual shares. In column (2), the additional regressor $\Delta\text{Equity}_{s,1990\to2002}$ is the target share of public equities between 1990 and 2002, constructed using target shares from the 1990 PENDAT survey when available and actual shares when not. Because target shares in 1990 are more sparsely populated, column (3) instead controls for the active change in public equities between 1990 and 2002, defined based on Calvet et al. (2009). The row "1990 method" indicates which of replacement rule is applied. Columns (4)-(5) add a vector of controls that is described fully in Section 3.2. These include (but are not limited to): the fraction of assets that are restricted by each state's constitution as of 1998 (Fraction of Restriction Assets$_{1998}$), the fraction of assets managed by pensions that require minimum qualifications on those making investment decisions (Financial Sophistication$_{1998}$), and the (asset-weighted) fraction of political board members as of 2002 based on Andonov et al. (2018). Fraction of Restriction Assets$_{1998}$ and Financial Sophistication$_{1998}$ are both based on the 1998 PENDAT survey when available and the 1996 survey otherwise. Robust standard errors are used in all regressions, and t-statistics are listed below point estimates in parentheses. ** indicates a p-value of 0.05 and * indicates a p-value of 0.10. See Section 3.2 for more complete details.

Table 6: Peer Effects

	Alternative-to-Risky Share					
	(1)	(2)	(3)	(4)	(5)	(6)
Peers' Alt-to-Risky Share	0.71**	0.65**	0.72**	0.75**		
	(3.30)	(3.76)	(3.31)	(3.49)		
× Established-CIO		0.00				
		(0.03)				
× Well-Funded			-0.10			
			(-0.84)			
× High-Performing				-0.19**		
				(-2.16)		
Lagged Peers' Alt-to-Risky Share					0.73**	
					(3.21)	
Non-Pension Peers' Alt-to-Risky Share						0.43**
						(2.26)
FE	(c,y,d)	(c,y,d)	(c,y,d)	(c,y,d)	(c,y,d)	(c,y)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Within-R^2	0.15	0.15	0.16	0.16	0.16	0.02
Total R^2	0.68	0.60	0.68	0.68	0.67	0.45
N	2,039	900	2,039	2,039	1,915	2,429

Notes: This table summarizes linear regressions of pension p's target alternative-to-risky share in year t on the target alternative-to-risky share of its peers. In columns (1)-(5), for a given pension, the target alternative-to-risky share of its peers is defined as a weighted average of all other pensions' target shares of alternatives, where weights are defined as the inverse of the distances between the given pension and each of its peers, respectively. In column (6), the alternative-to-risky share of each pension's peers is defined as the average (actual) alternative-to-risky share of unions, private-sector pensions, and endowments in its state. Columns (1)-(5) include consultant-by-time-by-census-division fixed effects and column (6) includes consultant-by-year fixed effects. All regressions include controls for funding ratio, asset hurdle rate, (log) size, required actuarial contributions relative to payroll, and administrative expenses relative to payroll. Columns (1) to (4) show the baseline relationship between the alternative-to-risky share for pension p in year t and its peers' share, as well as the interaction with several characteristics. In all cases, the regression includes both the characteristic and its interaction with peer target share of alternatives. The indicator variable "Established-CIO" in column (2) equals one if the pension has had the same CIO for at least five years. The regression in column (2) is estimated only using data after 2005. The indicator for "Well-Funded" in column (3) equals one if the pension is in the top quartile of funding for the entire sample. The indicator for "High-Performing" in column (4) equals one if the pension is in the top quartile of annual net-of-fee performance in a given year. Column (5) shows the regression of the alternative-to-risky share on the lagged peer target alternative share. Standard errors are clustered at the year and state levels, and t-statistics are listed below point estimates in parentheses. ** indicates a p-value of 0.05 and * indicates a p-value of 0.10.

Table 7: Risk-Seeking Explanations for the Rise in Alternatives

	ΔAlternative-to-Risky Share						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Δ GASB 25 Funding Ratio	-0.10				-0.08		
	(-1.16)				(-0.95)		
Δ BEA-Adjusted Funding Ratio		-0.06					
		(-0.28)					
Δ Liability Discount Rate			-1.02			-1.11	
			(-0.31)			(-0.30)	
Δ Fraction of Retired Members				0.13			0.13
				(0.76)			(0.72)
Aggregation	System	State	System	System	System	System	System
Constraint Control	No	No	No	No	Yes	Yes	Yes
Total R^2	0.01	0.00	0.00	0.00	0.03	0.02	0.02
N	118	47	117	115	116	115	113

Notes: This table shows regressions of changes in the target alternative-to-risky share on contemporaneous changes in several covariates. All changes are computed between 2002 and 2021. Risky investments are any holdings outside of fixed income and cash. Alternatives encompass investments in hedge funds, private equity, private debt, and real assets. GASB 25 funding ratios are based on liabilities that equal future promised benefits discounted at the assumed long-term rate of return for each plan. BEA-adjusted funding ratios instead discount future benefits using AAA-rated corporate borrowing rates. The liability discount rate is the one used for computing GASB 25 funding ratios and is also the system's return hurdle rate. The row labeled "Aggregation" specifies whether the regression is run at the system or state level. The row "Constraint Control" indicates whether controls are included for the average difference between each pension's actual and its target risky share between 2002 and 2020 (when both are available), as well as the change in their target cash share over the same period. Standard errors are clustered by state for regressions run at the system level and are robust for regressions run at the state level. t-statistics are listed below point estimates. ** indicates a p-value of 0.05 and * indicates a p-value of 0.10. See Section 2.1 for how we filter the data and Section 4.1 for more details on how we define the tightness of risk constraints.

Table 8: Spending Needs and Alternative Use

	Δ LE$_{s,2019\rightarrow2021}$	Δ FR$_{s,2019\rightarrow2022}$	$\Delta\Omega_{s,2019\rightarrow2022}$	$\Delta\Omega_{s,2019\rightarrow2023}$
	(1)	(2)	(3)	(4)
Covid Mortality Rate$_s$	-0.22**	-0.38**	-0.00	-0.00
	(-4.67)	(-2.24)	(-0.99)	(-0.67)
Constant	-1.05**	1.44**	0.01	0.02**
	(-22.91)	(7.47)	(1.17)	(2.62)
R^2	0.31	0.07	0.01	0.00
N	51	51	51	48

Notes: This table summarizes regressions of state-level changes in several variables on the average mortality rate from Covid-19 between 2020 and 2022, standardized to have a mean of zero and variance of one. The outcome variables across the four columns are as follows: (1) the change in age-65 life expectancy between 2019 and 2021 (ΔLE$_{s,2019\rightarrow2021}$); (2) the change in the percent of retired members from 2019 to 2022 (ΔFR$_{s,2019\rightarrow2022}$); (3) the change in the alternative-to-risky share from 2019 to 2022 ($\Delta\Omega_{2019\rightarrow2022}$); and (4) the change in the target alternative-to-risky share from 2019 to 2023 ($\Delta\Omega_{2019\rightarrow2023}$). Mortality and life expectancy data are sourced from the Centers for Disease Control and Prevention (CDC). Changes in life expectancy are computed only through 2021, as data for 2022 are not yet available. t-statistics are listed below point estimates and are based on heteroskedasticity-robust standard errors. ** indicates a p-value of 0.05 and * indicates a p-value of 0.10.

Table 9: Simulation Exercise: Parametrization

Panel A: Randomly drawn parameters

Parameter	Definition	Sampling Distribution
μ_E	Expected excess return of public equities	$U[0.02, 0.07]$
σ_E^2	Variance of public equities	$U[0.02, 0.09]$
α	Alpha of alternatives	$U[0, 0.05]$
β	Beta of alternatives	$U[0, 1.5]$

Panel B: Inferred Parameters

Parameter	Definition	Method
μ_A	Expected excess return of alternatives	$\mu_A = \alpha + \beta\mu_E$
σ_A^2	Variance of alternatives	$\sigma_A^2 = \beta^2\sigma_E^2 + \sigma_\eta^2$
σ_{AE}	Covariance of public equities and alternatives	$\sigma_{AE} = \beta\sigma_E^2$
σ_η^2	Idiosyncratic variance of alternatives	Chosen to match initial risky portfolio composition (Eq. (1) and (2))
ω_f^{min}	Minimum constraint on riskless asset	Based on observed fixed income share in 2021

Panel C: Sample Size	Sample size
Admissible beliefs S^*:	$100,000$
Risk aversion able to explain $\Delta\omega_A^*$:	379
Risk aversion unable to explain $\Delta\omega_A^*$:	$99,621$
Pensions shift to public equities when constrained	$99,137$
Pensions shift to alternatives when constrained, but γ_2 binds at 1	484

Panel A of this table summarizes the distributions from which beliefs are drawn in Section 5.1, where we simulate a decline in risk aversion paired with a binding portfolio constraint. Panel B summarizes how the remaining parameters are backed out from the model in Section 2.3. Panel C provides a breakdown of when a decline in risk aversion is able to generate the observed aggregate change in the target alternative-to-risky share from 2001 to 2021. See Section 5.1 for more details.